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1. Introduction

The Hidden Markov Model (HMM) is a well-studied statistical model that has evoked great interest and remarkable results in the past decades in both research and practice. In simple words, HMM is a stochastic process which underlies a hidden Markov process but can only be observed by probabilistic signals emitted by states of this hidden process. Originally, it was first studied by Baum and Petrie [3] as *probabilistic functions of Markov chains* and gained significance during the 1980s and 1990s as application in speech recognition, see Rabiner [31]. During this time, HMMs dominated automated speech recognition research and commercial application, forming the backbone of early voice-controlled interfaces. In general, HMMs are applied for tasks that require decoding, recognition and classification like DNA analysis and gene recognition [27, 34] or behavior detection [6]. Recently, Makonin, Popowich, Bajić, Gill, and Bartram [28] used HMM to disaggregate electric power load profiles and identify the usage of single electric devices. In the last years, however, machine learning techniques like recurrent neural networks or transformers have superseded HMMs especially in large-scale tasks of decoding and classification. Nevertheless, HMM still remains a valuable tool in many applications due to its interpretability, mathematical tractability, and efficiency in low-data settings.

In this thesis, our aim is to investigate another possible application of HMM, namely forecasting. In the existing literature, there have been only a few single studies conducted using HMM as a forecasting method. Erlwein, Benth, and Mamon [10] developed a HMM-driven method to model the dynamics of the one-step-a-head prediction for the electrical spot price market for which 2- and 3-states Markov processes are trained to estimate certain dynamic parameters. In Tian and Shen [35], different Markovian models including 4 HMMs are compared to predict the probability of the binary event of US recession. Ghasvarian Jahromi, Gharavian, and Mahdiani [11] uses a common approach from speech recognition in which multiple HMM model are trained on wind speed data and applied the currently most fitting model for prediction. Stock market predictions using HMM were studied in Gupta and Dhingra [15] and Hassan and Nath [18], while Khadr [25] developed a HMM based drought forecasting process. However, to the best of the author’s knowledge, the literature of HMM in forecasting has several shortcomings. Foremost, there exists no general theory on HMM as a forecasting method, which is commonly acknowledged and based upon. The theoretic approach of many papers are often tightly tailored to their specific model but actually rely on similar ideas, while only a few take completely different approaches. This makes it not only difficult to understand in many cases how the forecasts were actually derived, but it also blurs the distinction between already existing methods and novelties in this field.

Secondly, all reviewed papers apply small-scale HMMs (less than 10 hidden states) with predefined hyperparameter settings. While these states are often interpreted by practical

groundings (like recession vs economic growth), it is not given that these assumptions are actually the optimal hyperparameter settings. It should be standard procedure to conduct a hyperparameter tuning like in classical machine learning methodology, but also to consider much bigger models which have become feasible with nowadays computer. Thirdly, most literature (with exception of [11]) derive deterministic or point forecasts from their HMM forecasting method which is achieved by most often taking the most probable event as a prediction. However, this approach strips away the HMM method from a significant part of its provided information, since it considers the probabilities for all possible outcomes due to its stochastic design. Taking full advantage of this stochastic design results in probabilistic forecasting. Probabilistic forecasts, for example predicting the distribution of a future event, have been applied for many years exclusively in very niche applications like meteorology. However in the last decade, it has risen great interest in many fields due to its capability to measure the uncertainty of forecasting. One example is the area of finance and economics in which probabilistic forecasting offers a great tool for risk assessment. One challenge of applying probabilistic forecasting methods is often the more advanced design and greater computational effort in comparison to their deterministic counterparts. Precisely for this reason, the application of HMM as probabilistic forecasting method offers a great opportunity. Not only does it have a straight forward design that is easy to understand, implement and efficiently train, but also provides directly probabilistic forecasts. Further, in comparison to many black-box machine learning methods, HMMs are described in closed mathematically form and its interpretable parameters offer insight into its decision-making. These benefits can allow the HMM forecasting model to be employed for critical systems in which the model's decision has to be understood or be included in theoretical frameworks for further studies like stochastic optimization systems.

This thesis contributes in filling the three research gaps described above. We derive a general theoretic framework for the application of HMM as a forecasting method, which builds upon and uses similar approaches as the standard HMM work of Rabiner [31]. Based on this general method, we deploy HMM in a use-case of probabilistic forecasting achieving better performances of standard probabilistic forecasting methods. For the training of the HMM forecasting model, a thorough methodology is developed to investigate the change of hyperparameter settings, sensitivity and predictive behavior. Here, we study in detail the standard hyperparameters of HMM and in addition also consider different types of discretization and historic window length which is a novelty for HMM. The studied use-case of this work is the short-term, probabilistic forecast of the electrical power load for single households. In detail, we will predict the distribution of a household's electrical power demand 15 minutes ahead using a novel HMM forecasting method. Electrical power load forecasting is an essential task for energy providers and power grid operators to guarantee energy supply and security. In times of energy transition, the rising demand of electricity and increasing level of volatile renewable energy are determined to challenge the current electrical power systems. For this reason, both research and companies invest large efforts to find new solutions and innovations to increase the efficiency of the existing systems. The digitalisation of the power systems in the last years,

in particular the widespread roll-out of smart meters, contributed to a great increase of available data, literature and new forecasting methods in the field of low voltage energy systems. Haben, Arora, Giasemidis, Voss, and Vukadinović Greetham [16] conducted an encompassing review of low voltage load forecasting, identifying multiple research gap and challenges, and provides recommendations for future work. One key take away from this review is that point forecasts are often not sufficient due to the high volatility of demand on the level of low voltage. A similar argument was stated within the *global energy forecasting competitions* (GEFCom), see Hong, Pinson, and Fan [20], Hong, Pinson, Fan, Zareipour, Troccoli, and Hyndman [21], and Hong, Xie, and Black [22]. While the first GEFCom2012 addressed point forecasts, the following competitions GEFCom2014 and GEFCom2017 exclusively targeted probabilistic forecasts.

Our use-case of probabilistic forecasting the load of single households is potentially relevant for two parties. Home owners are confronted with an increase in electric demand in the upcoming future due to the electrification of transport (E-mobility) and heating (heat pumps). Using home energy management systems (HEMS) can optimize the operation of the home system via different components (PV electricity production, battery storage management, flexible demand, dynamic pricing) for which good demand predictions are valuable. For example, HEMS can utilize probabilistic forecasts to evaluate the risk of crossing a certain power threshold¹ and act accordingly. Essential challenges for home owners to implement forecasting methods are dominantly lack of available data and computational limited resources. The proposed HMM forecasting method fits to these conditions since it is trained only on historical load data without additional information (like eg. weather data) and efficiently due to its relatively simple mathematical design and algorithms. The second potential target group are distribution grid operators. For a robust operation of any electrical power grid, accurate forecasts are essential to avoid grid congestion and expensive re-dispatching. By aggregating multiple household's forecasts, the implementation of a hierarchical bottom-up forecasting model is made possible by the efficiency of the proposed HMM forecasting method.

The choice of applying the proposed HMM forecasting method for probabilistic forecasting the electrical power load of households is well justified by following summarizing arguments. The novel approach of HMM forecasting method offers great opportunities in the field of energy forecasting, especially probabilistic load forecasting, because its stochastic design provides an efficient alternative to existing probabilistic forecasting methods. Further, probabilistic forecasts show great benefits over point estimations for high volatile data such as household's electric demand. Lastly, the proposed HMM forecasting method satisfies the requirements for potential applications of household's electrical power forecasting.

Following bullet points highlight the contribution of this thesis to current research:

- Development of a general theoretic framework for HMM based forecasting
- Deployment of a novel probabilistic HMM forecasting method on households electri-

¹In Austrian's upcoming law (ElWG-Gesetz), home owners with PV production will be penalized by a fee, if their current supply into the grid exceeds 7kW.

cal power load, offering an efficient alternative to existing probabilistic forecasting methods

- Detailed analysis of the prediction model's hyperparameter and behavior, including a novel discretization method for HMM

The presented thesis is structured as follows. Chapter 2 covers the theoretic foundations required including the necessary theory of HMM and probabilistic forecasting. Notably in section 2.1.3, the theoretic framework for HMM based forecasting and a forecasting algorithm are introduced. Chapter 3 discusses the experiment design and methodology of the use-case of the HMM forecasting method for probabilistic load prediction. In chapter 4 and 5, the results of the hyperparameter tuning and model analysis are presented respectively. Benchmarks are conducted in chapter 6 in order to compare the proposed HMM forecasting method to other state-of-the-art probabilistic load forecasting methods.

2. Theoretic Foundations

2.1. Hidden Markov Models

HMM has proved to be a valuable extension of the general Markov processes because of its capability to model signals and other volatile data. This has been shown in its wide application for speech recognition. The most popular paper and standard reference in this field is Rabiner [31] summarizing the main challenges of HMM, presenting the most important results and algorithms and gives an introduction to speech recognition applications. Other HMM related work builds upon the results and theory discussed in that paper. In this thesis, we exploit a relatively unexplored application of HMM as a forecasting method, for which, however, also already existing algorithms are required. For this reason, we give an encompassing introduction to the theory and present existing algorithms to solve important problems regarding HMM. In the last section, we derive the general theory for HMM forecasting and formulate an efficient HMM forecasting algorithm.

2.1.1. HMM Foundations

Starting with the notion of HMM, there are N possible hidden states forming the *state space* S and M different observation values forming the *observation space* O ,

$$\begin{aligned} S &= \{s_1, s_2, \dots, s_N\} \\ O &= \{o_1, o_2, \dots, o_M\}. \end{aligned}$$

We restrict the discussion to discrete HMM with finite state and observation space, thus $N, M \in \mathbb{N}$. Denote the sets of indexes by $I_S = \{1, \dots, N\}$ and $I_O = \{1, \dots, M\}$. Let $T \in \mathbb{N}$ be the time horizon and its corresponding index set $I_T = \{1, \dots, T\}$. The HMM consists of two dependent stochastic processes, a hidden and an observable one. For each time instance, the hidden process $\{Z_t, t \in I_T\}$ takes on values in the state space, while the observable process $\{X_t, t \in I_T\}$ is located in the observation space,

$$\begin{aligned} (Z_1, Z_2, \dots, Z_T) &\in S^T \\ (X_1, X_2, \dots, X_T) &\in O^T. \end{aligned}$$

To distinguish between random variables and the event of random variables taking fixed values, we introduce the following compact notation. The event of Z_t taking a value is denoted by $\dot{Z}_t$,

$$\dot{Z}_t = \{Z_t = s_{i_t}\},$$

for a fixed $i_t \in I_S$ that describes the state index at time t . All events until time instance t summed up by the event vector $\dot{\mathbf{Z}}_t$,

$$\dot{\mathbf{Z}}_t = (\dot{Z}_1, \dot{Z}_2, \dots, \dot{Z}_t).$$

Analogously for the observable process, the fixed $i_t \in I_S$ describes the observation index at time t ,

$$\dot{X}_t = \{X_t = o_{k_t}\},$$

In addition to $\dot{\mathbf{X}}_t$, the vector $\dot{\mathbf{X}}'_t$ describes all occurred events after time instance t (including time instance t),

$$\begin{aligned}\dot{\mathbf{X}}_t &= (\dot{X}_1, \dot{X}_2, \dots, \dot{X}_t) \\ \dot{\mathbf{X}}'_t &= (\dot{X}_t, \dot{X}_{t+1}, \dots, \dot{X}_{T-1}, \dot{X}_T).\end{aligned}$$

The HMM is entirely characterized through the following three properties.

(A1) Markov property:

$$\mathcal{P}(\dot{Z}_{t+1} | \dot{\mathbf{Z}}_t) = \mathcal{P}(\dot{Z}_{t+1} | \dot{Z}_t)$$

(A2) Homogeneity:

$$\mathcal{P}(Z_{t+1} = s_j | Z_t = s_i) = \mathcal{P}(Z_t = s_j | Z_{t-1} = s_i) = \dots = \mathcal{P}(Z_2 = s_j | Z_1 = s_i)$$

(A3) Independent observation assumption:

$$\mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{t-1}, \dot{\mathbf{X}}'_{t+1}, \dot{\mathbf{Z}}_T) = \mathcal{P}(\dot{X}_t | \dot{Z}_t)$$

The properties **A1** and **A2** determine the process Z to be a homogeneous Markov chain. Thus, a future event depends on the past events only through the present event. The third property **A3** implies that the observation is influenced only by the current hidden state. Due to these properties, a HMM can be parameterized by stochastic matrices and vectors similar to the transition matrix of standard Markov chains. The hidden Markov process $\{Z_t, t \in I_T\}$ is determined by the $N \times N$ -dimensional *transition matrix* $\mathbb{A}$ with elements a_{ij} ,

$$a_{ij} = \mathcal{P}(Z_t = s_j | Z_{t-1} = s_i) \quad \text{for } i, j = 1, \dots, N,$$

and by the N -dimensional *initial distribution vector* π with

$$\pi_i = \mathcal{P}(Z_1 = s_i) \quad \text{for } i = 1, \dots, N.$$

The $N \times M$ -dimensional *observation matrix* $\mathbb{B}$ describes the emission probabilities of the observation process $\{X_t, t \in I_T\}$ with elements b_{ik} ,

$$b_{ik} = \mathcal{P}(X_t = o_k | Z_t = s_i) \quad \text{for } i = 1, \dots, N \text{ and } k = 1, \dots, M.$$

These three parameters fully describe a finite HMM, summarized by $\lambda = (\mathbb{A}, \mathbb{B}, \pi)$.

2.1.2. Algorithms

The three strong properties [A1](#), [A2](#), [A3](#) allow us to compute HMM related problems very efficiently. Before we present common algorithms, we first identify the three main questions regarding HMM formulated by Rabiner [\[31\]](#):

1. **Evaluation:**

Given an HMM and observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$, what is the probability that this sequence occurred,

$$\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})?$$

2. **Decoding:**

Given a HMM and observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$, what is the most probable state sequence $\dot{\mathbf{Z}}_{\mathbf{T}}$ which emits the given observations,

$$\arg \max_{\dot{\mathbf{Z}}_{\mathbf{T}}} \mathcal{P}(\dot{\mathbf{Z}}_{\mathbf{T}} | \dot{\mathbf{X}}_{\mathbf{T}})?$$

3. **Training:**

Given observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$, what are the optimal parameters λ of the HMM to describe the given observations,

$$\arg \max_{\lambda} \mathcal{P}_{\lambda}(\dot{\mathbf{X}}_{\mathbf{T}})?$$

Due to the three properties of HMMs, all of these quantities can be efficiently computed. In this section, we discuss solutions and algorithms for the problem of evaluation and training, which is relevant for the application of HMM as a forecasting tool. The question of decoding can also be relevant for the analysis of existing forecasting models to develop a deeper understanding of the underlying mechanisms, which however we do not discuss further in this thesis.

To show the computational complexity of the above problems, we first make an exemplary naive approach of calculating the evaluation problem. Given $\dot{\mathbf{X}}_{\mathbf{T}}$, we reformulate the desired probability using the conditional probability over all possible hidden state sequences $\dot{\mathbf{Z}}_{\mathcal{T}}$,

$$\begin{aligned} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) &= \sum_{\dot{\mathbf{Z}}_{\mathbf{T}} \in \dot{\mathcal{Z}}_{\mathcal{T}}} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}, \dot{\mathbf{Z}}_{\mathbf{T}}) \\ &= \sum_{\dot{\mathbf{Z}}_{\mathbf{T}} \in \dot{\mathcal{Z}}_{\mathcal{T}}} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}} | \dot{\mathbf{Z}}_{\mathbf{T}}) \mathcal{P}(\dot{\mathbf{Z}}_{\mathbf{T}}). \\ &= \sum_{\dot{\mathbf{Z}}_{\mathbf{T}} \in \dot{\mathcal{Z}}_{\mathcal{T}}} \left(\prod_{t=1}^T \mathcal{P}(\dot{X}_T | \dot{Z}_T) \right) \left(\mathcal{P}(\dot{Z}_1) \prod_{t=2}^T P(\dot{Z}_t | \dot{Z}_{t-1}) \right) \\ &= \sum_{\dot{\mathbf{Z}}_{\mathbf{T}} \in \dot{\mathcal{Z}}_{\mathcal{T}}} \left(\prod_{t=1}^T b_{i_t k_t} \right) \left(\pi_{i_1} \prod_{t=2}^T a_{i_{t-1} i_t} \right). \end{aligned}$$

The set of all possible hidden state sequences is defined formally by

$$\mathcal{Z}_T = \{(Z_t = s_{i_t})_{t \in \{1, \dots, T\}} \text{ for } (i_t)_{t \in \{1, \dots, T\}} \in I_S^T\}.$$

Since I_S has N elements, there are N^T different possible hidden state sequences. Thus, evaluating $\mathcal{P}(\dot{\mathbf{X}}_T)$ has a non-polynomial complexity of order $\mathcal{O}(N^T \cdot 2T)$ and for larger models or longer time horizon reaches quickly the limits of computational feasibility. However, due to its Markovian behavior, the evaluation of $\mathcal{P}(\dot{\mathbf{X}}_T)$ can be done more efficiently by building up an iterative procedure based on its three properties. This is called the *forward algorithm* presented in the following.

We define the quantity $\alpha_t(i)$ as

$$\alpha_t(i) = \mathcal{P}(\dot{\mathbf{X}}_t, Z_t = s_i), \quad \text{for } i \in I_S, t \in \{1, \dots, T\}.$$

For $t \geq 2$, the joint probability $\alpha_t(j)$ can be expressed by its predecessors $\alpha_{t-1}(\cdot)$ and the parameters of the given HMM,

$$\begin{aligned} \alpha_t(j) &= \mathcal{P}(\dot{\mathbf{X}}_t, Z_t = s_j) \\ &= \sum_{i \in I_S} \mathcal{P}(\dot{\mathbf{X}}_t, Z_t = s_j, Z_{t-1} = s_i) \\ &= \sum_{i \in I_S} \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{t-1}, Z_t = s_j, Z_{t-1} = s_i) \mathcal{P}(Z_t = s_j | \dot{\mathbf{X}}_{t-1}, Z_{t-1} = s_i) \\ &\quad \cdot \mathcal{P}(\dot{\mathbf{X}}_{t-1}, Z_{t-1} = s_i) \\ &= \sum_{i \in I_S} \mathcal{P}(\dot{X}_t | Z_t = s_j) \mathcal{P}(Z_t = s_j | Z_{t-1} = s_i) \mathcal{P}(\dot{\mathbf{X}}_{t-1}, Z_{t-1} = s_i) \\ &= \sum_{j \in I_S} b_{jk_t} \cdot a_{ij} \cdot \alpha_{t-1}(i). \end{aligned}$$

The computation of the first joint probabilities $\alpha_1(\cdot)$ requires the initial distribution π ,

$$\alpha_1(i) = \mathcal{P}(\dot{X}_1, Z_1 = s_i) = \mathcal{P}(\dot{X}_1 | Z_1 = s_i) \mathcal{P}(Z_1 = s_i) = b_{ik_1} \cdot \pi_i.$$

Using the quantities $\alpha_T(\cdot)$, the probability targeted in the evaluation problem is calculated by

$$\mathcal{P}(\dot{\mathbf{X}}_T) = \sum_{i \in I_S} \mathcal{P}(\dot{\mathbf{X}}_T, Z_T = s_i) = \sum_{i \in I_S} \alpha_T(i).$$

This is the standard forward algorithm, discussed in [31], and has a computational complexity of order $\mathcal{O}(N^2 \cdot 3T)$, which is significantly lower than that of the naive approach. However, this algorithm also has its drawbacks in practice. If we look closely at the definition of the joint probability $\mathcal{P}(\dot{\mathbf{X}}_T)$, it can be observed that with increasing time horizon T this probability is becoming smaller and in regular cases converges to 0. It implies that with a significantly large time horizon, this quantity will be smaller than any

for a computer representable number and thus not feasible to compute. For this reason, we present a version of the forward algorithm adapted to conditional probabilities which do not vanish with increasing T and allows for the computation of the log-likelihood of the evaluation problem.

We define two quantities $\bar{\alpha}$ and $\tilde{\alpha}$ similar to the original α ,

$$\bar{\alpha}_t(i) = \mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}-1})$$

and

$$\tilde{\alpha}_t(i) = \mathcal{P}(Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}}),$$

for $i \in I_S$ and $t \in \{1, \dots, T\}$. On the one hand, $\bar{\alpha}_t(i)$ is computed by the predecessors $\tilde{\alpha}_{t-1}(\cdot)$,

$$\begin{aligned} \bar{\alpha}_t(i) &= \mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}-1}) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_t, Z_t = s_i, Z_{t-1} = s_j | \dot{\mathbf{X}}_{\mathbf{t}-1}) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}-1}, Z_{t-1} = s_j) \mathcal{P}(Z_{t-1} = s_j | \dot{\mathbf{X}}_{\mathbf{t}-1}) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_t | Z_t = s_i) \mathcal{P}(Z_t = s_i | Z_{t-1} = s_j) \mathcal{P}(Z_{t-1} = s_j | \dot{\mathbf{X}}_{\mathbf{t}-1}) \\ &= \sum_{j \in I_S} b_{ikt} \cdot a_{ji} \cdot \tilde{\alpha}_{t-1}(j). \end{aligned}$$

Considering the fact of

$$\begin{aligned} \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-t}) &= \sum_{i \in I_S} \mathcal{P}(Z_t = s_i, \dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-t}) \\ &= \sum_{i \in I_S} \bar{\alpha}_t(i), \end{aligned}$$

we can express on the other hand $\tilde{\alpha}_t(i)$ by $\bar{\alpha}_t(\cdot)$,

$$\begin{aligned} \tilde{\alpha}_t(i) &= \mathcal{P}(Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}}) \\ &= \frac{\mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}-t})}{\mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-t})} \\ &= \frac{\bar{\alpha}_t(i)}{\sum_{j \in I_S} \bar{\alpha}_t(j)}. \end{aligned}$$

As mentioned, the probability $\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})$ induces underflow errors. However, the log-likelihood of the evaluation problem, $\log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})$ lies generally within the computationally feasible range. The log-likelihood is calculated by splitting up the original problem into

subproblems,

$$\log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) = \log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) = \log \prod_{t=1}^T \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1}) = \sum_{t=1}^T \log \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1}),$$

and using the fact that $\mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1}) = \sum_{j \in I_S} \bar{\alpha}_j(t)$ as shown above. Additionally, consider the following relation between $\tilde{\alpha}$ and α ,

$$\begin{aligned} \tilde{\alpha}_t(i) &= \mathcal{P}(Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}}) \\ &= \frac{\mathcal{P}(Z_t = s_i, \dot{\mathbf{X}}_{\mathbf{t}})}{\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}})} \\ &= \frac{\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}}, Z_t = s_i)}{\sum_{j \in I_S} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}}, Z_t = s_j)} \\ &= \frac{\alpha_t(i)}{\sum_{j \in I_S} \alpha_j(t)}. \end{aligned}$$

This shows that $\tilde{\alpha}_t(i) = c_t \cdot \alpha_t(i)$ with the constant factor $c_t = \frac{1}{\sum_{j \in I_S} \alpha_j(t)}$. Thus, $\tilde{\alpha}_t$ is a scaled version of α_t , which becomes relevant in the training problem.

Adapted forward algorithm

1. *Initialization:* ($t = 1$)

$$\begin{aligned} \bar{\alpha}_1(i) &= b_{ik_1} \cdot \pi_i \quad \text{for } i \in I_S \\ \mathcal{P}(\dot{X}_1) &= \sum_{j \in I_S} \bar{\alpha}_1(j) \\ \tilde{\alpha}_1(i) &= \bar{\alpha}_1(i) / \mathcal{P}(\dot{X}_1) \quad \text{for } i \in I_S \end{aligned}$$
2. *Iteration:* ($t = 2, \dots, T$)

$$\begin{aligned} \bar{\alpha}_t(i) &= \sum_{j \in I_S} b_{ikt} \cdot a_{ji} \cdot \bar{\alpha}_{t-1}(j) \quad \text{for } i \in I_S \\ \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1}) &= \sum_{j \in I_S} \bar{\alpha}_j(t) \\ \tilde{\alpha}_t(i) &= \bar{\alpha}_t(i) / \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1}) \quad \text{for } i \in I_S \end{aligned}$$
3. *Termination:*

$$\log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) = \sum_{t=1}^T \log \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1})$$

return $\log P(\dot{\mathbf{X}}_{\mathbf{T}})$

In addition, the evaluation can also be computed with the so-called *backward algorithm*. It utilizes the HMM properties similar as the forward algorithm but iterates from the other temporal direction, thus backwards. Both of these algorithms are needed later on to solve the training problem.

Let $\beta_t(i)$ be the following quantity for $t \in \{1, \dots, T-1\}$ and $i \in I_S$,

$$\beta_t(i) = \mathcal{P}(\dot{\mathbf{X}}'_{\mathbf{t}+1} | Z_t = s_i).$$

The probabilities $\beta_t(\cdot)$ are described recursively and initiated by setting $\beta_T(\cdot)$ to one,

$$\beta_T(i) = 1.$$

Then,

$$\begin{aligned} \beta_t(i) &= \mathcal{P}(\dot{X}_{t+1}, \dot{\mathbf{X}}'_{t+2} | Z_t = s_i) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_{t+1}, \dot{\mathbf{X}}'_{t+2} | Z_t = s_i, Z_{t+1} = s_j) \mathcal{P}(Z_{t+1} = s_j | Z_t = s_i) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_{t+1} | Z_{t+1} = s_j) \mathcal{P}(\dot{\mathbf{X}}'_{t+2} | Z_{t+1} = s_j) \mathcal{P}(Z_{t+1} = s_j | Z_t = s_i) \\ &= \sum_{j \in I_S} b_{jk_{t+1}} \cdot \beta_{t+1}(j) \cdot a_{ij}. \end{aligned}$$

Since $\dot{\mathbf{X}}_T = \dot{\mathbf{X}}'_1$, the desired probability of the evaluation problem is then computed by,

$$\begin{aligned} \mathcal{P}(\dot{\mathbf{X}}_T) &= \mathcal{P}(\dot{\mathbf{X}}'_1) \\ &= \sum_{i \in I_S} \mathcal{P}(\dot{\mathbf{X}}'_1 | Z_1 = s_i) \mathcal{P}(Z_1 = s_i) \\ &= \sum_{i \in I_S} \mathcal{P}(\dot{X}_1 | Z_1 = s_i) \mathcal{P}(\dot{\mathbf{X}}'_2 | Z_1 = s_i) \mathcal{P}(Z_1 = s_i) \\ &= \sum_{i \in I_S} b_{ik_1} \cdot \beta_1(j) \cdot \pi_i. \end{aligned}$$

Similar to the forward algorithm, the standard version of the backward algorithm as described above induces numerical difficulties in practice and therefore, demands an adaptation. To avoid infeasibility, $\beta_t(\cdot)$ is scaled with the constant $c_t = \sum_{j \in I_S} \beta_t(j)$ for $t \in \{1, \dots, T-1\}$ and $c_T = 1$. summarized by $\tilde{\beta}_t(\cdot)$,

$$\tilde{\beta}_t(i) = \frac{\beta_t(i)}{c_t} = \frac{\mathcal{P}(\dot{\mathbf{X}}'_{t+1} | Z_t = s_i)}{\sum_{j \in I_S} \mathcal{P}(\dot{\mathbf{X}}'_{t+1} | Z_t = s_j)}$$

This scaling implies that

$$\sum_{i \in I_S} \tilde{\beta}_t(i) = 1,$$

and thus assures feasibility for all $\tilde{\beta}_t(i)$. The factor c_t has no meaningful interpretation as the scaling factor of the adapted forward algorithm and also cannot be directly computed.

However, due to the above result, the ratio c_t/c_{t+1} is calculated as follows,

$$\begin{aligned}
 1 &= \sum_{i \in I_S} \tilde{\beta}_t(i) \\
 &= \sum_{i \in I_S} \frac{\beta_t(i)}{c_t} \\
 &= \sum_{i \in I_S} \frac{\sum_{j \in I_S} b_{jk_{t+1}} \cdot a_{ij} \cdot \beta_{t+1}(j)}{c_t} \\
 &= \sum_{i \in I_S} \frac{\sum_{j \in I_S} b_{jk_{t+1}} \cdot a_{ij} \cdot \tilde{\beta}_{t+1}(j) \cdot c_{t+1}}{c_t} \\
 &= \frac{c_{t+1}}{c_t} \sum_{i \in I_S} \sum_{j \in I_S} b_{jk_{t+1}} \cdot a_{ij} \cdot \tilde{\beta}_{t+1}(j).
 \end{aligned}$$

Hence,

$$\frac{c_t}{c_{t+1}} = \sum_{i \in I_S} \sum_{j \in I_S} b_{jk_{t+1}} \cdot a_{ij} \cdot \tilde{\beta}_{t+1}(j).$$

The log-likelihood is then derived by

$$\begin{aligned}
 \log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) &= \log \left(\sum_{i \in I_S} b_{ik_1} \cdot \pi_i \cdot \beta_1(j) \right) \\
 &= \log \left(\sum_{i \in I_S} b_{ik_1} \cdot \pi_i \cdot \tilde{\beta}_1(j) \cdot c_1 \right) \\
 &= \log \left(\sum_{i \in I_S} b_{ik_1} \cdot \pi_i \cdot \tilde{\beta}_1(j) \cdot \prod_{t=1}^{T-1} \frac{c_t}{c_{t+1}} \right) \\
 &= \log \left(\sum_{i \in I_S} b_{ik_1} \cdot \pi_i \cdot \tilde{\beta}_1(j) \right) + \sum_{t=1}^{T-1} \log \frac{c_t}{c_{t+1}}.
 \end{aligned}$$

Adapted backward algorithm

1. *Initialization:* ($t = T$)
 $\tilde{\beta}_T(i) = 1$
 $c_T = 1$
 2. *Iteration:* ($t = T - 1, \dots, 1$)
 $\frac{c_t}{c_{t+1}} = \sum_{i \in I_S} \sum_{j \in I_S} b_{jk_{t+1}} \cdot a_{ij} \cdot \tilde{\beta}_{t+1}(j)$
 $\beta_t(i) = \sum_{j \in I_S} b_{jk_{t+1}} \cdot a_{ij} \cdot \tilde{\beta}_{t+1}(j)$
 3. *Termination:*
 $\log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) = \log \left(\sum_{i \in I_S} b_{ik_1} \cdot \pi_i \cdot \tilde{\beta}_1(j) \right) + \sum_{t=1}^{T-1} \log \frac{c_t}{c_{t+1}}$
- return* $\log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})$

Turning now to the more advanced problem of training HMMs, we introduce the *Baum-Welch algorithm*. The objective is to find the optimal parameters of an HMM to fit a given observation sequence. The Baum-Welch algorithm is a special case of the expectation maximization (EM) algorithm, which is an iterative approach to find the local maxima of the likelihood for the parameter settings of a statistical model. Each iteration is divided into two steps, the expectation and the maximization step. While in the expectation step, the current expectational values under fixed parameters are computed, these expectational values are then used in the maximization step to re-estimate the next parameters. The reader is referred to Dempster, Laird, and Rubin [9] for a detailed discussion of EM algorithms and to Baum [2] for a formal derivation of the Baum-Welch algorithm. In this work, we illustrate the main ideas of Baum-Welch and give an intuitional approach. Let $\xi_t(i, j)$ be the conditional probability of being in the hidden state s_i at the time instance t and transitioning to state s_j under the given observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$,

$$\xi_t(i, j) = \mathcal{P}(Z_t = s_i, Z_{t+1} = s_j \mid \dot{\mathbf{X}}_{\mathbf{T}}).$$

Due to the fact of

$$\begin{aligned} \mathcal{P}(Z_t = s_i, Z_{t+1} = s_j, \dot{\mathbf{X}}_{\mathbf{T}}) &= \\ &= \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}} \mid Z_t = s_i, Z_{t+1} = s_j) \mathcal{P}(Z_t = s_i, Z_{t+1} = s_j) = \\ &= \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}} \mid Z_t = s_i) \mathcal{P}(\dot{\mathbf{X}}'_{\mathbf{t}+1} \mid Z_{t+1} = s_j) \mathcal{P}(Z_{t+1} = s_j \mid Z_t = s_i) \mathcal{P}(Z_t = s_i) \\ &= \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}}, Z_t = s_i) \mathcal{P}(\dot{\mathbf{X}}'_{\mathbf{t}+1} \mid Z_{t+1} = s_j) \mathcal{P}(\dot{X}_{t+1} \mid Z_{t+1} = s_j) \mathcal{P}(Z_{t+1} = s_j \mid Z_t = s_i) \\ &= \alpha_t(i) \cdot \beta_{t+1}(j) \cdot b_{jk_{t+1}} \cdot a_{ij}, \end{aligned}$$

we can express $\xi_t(i, j)$ as the following,

$$\begin{aligned}\xi_t(i, j) &= \frac{\mathcal{P}(Z_t = s_i, Z_{t+1} = s_j, \dot{\mathbf{X}}_{\mathbf{T}})}{\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})} \\ &= \frac{\mathcal{P}(Z_t = s_i, Z_{t+1} = s_j, \dot{\mathbf{X}}_{\mathbf{T}})}{\sum_{i' \in I_S} \sum_{j' \in I_S} \mathcal{P}(Z_t = s_{i'}, Z_{t+1} = s_{j'}, \dot{\mathbf{X}}_{\mathbf{T}})} \\ &= \frac{\alpha_t(i) \cdot a_{ij} \cdot b_{jk_{t+1}} \cdot \beta_{t+1}(j)}{\sum_{i' \in I_S} \sum_{j' \in I_S} \alpha_t(i') \cdot a_{i'j'} \cdot b_{j'k_{t+1}} \cdot \beta_{t+1}(j')}.\end{aligned}$$

Where the quantities $\alpha_t(i)$ and $\beta_t(j)$ are used from the forward and backward algorithm. Since $\alpha_t(\cdot)$ and $\beta_t(\cdot)$ are both used in the nominator as well as the denominator, they can be exchanged by its scaled versions $\tilde{\alpha}_t(\cdot)$ and $\tilde{\beta}_t(\cdot)$.

Further, consider the probability of being in state i during the time instance t given $\dot{\mathbf{X}}_{\mathbf{T}}$, thus¹

$$\gamma_t(i) = \mathcal{P}(Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{T}}) = \sum_{j \in I_S} \mathcal{P}(Z_t = s_i, Z_{t+1} = s_j | \dot{\mathbf{X}}_{\mathbf{T}}) = \sum_{j \in I_S} \xi_t(i, j).$$

Given the quantities $\xi_t(\cdot, \cdot)$ and $\gamma_t(\cdot)$, useful information on the hidden process is available. The expected number of transitions from state s_i to state s_j is equivalent to the sum of all $\xi_t(i, j)$ over $t \in \{1, \dots, T-1\}$,

$$\mathbb{E}(\#transitions \text{ form } s_i \text{ to } s_j) = \sum_{t=1}^{T-1} \xi_t(i, j).$$

The expected number of visits in state s_i is described by the sum of all $\gamma_t(i)$ over $t \in \{1, \dots, T\}$,

$$\mathbb{E}(\#visits \text{ in } s_i) = \sum_{t=1}^T \gamma_t(i).$$

Similarly, the expected number of transitions from state s_i is the sum of all $\gamma_t(i)$ over $t \in \{1, \dots, T-1\}$,

$$\mathbb{E}(\#transitions \text{ from } s_i) = \sum_{t=1}^{T-1} \gamma_t(i).$$

The above equations represent the expectation step of the Baum-Welch algorithm. In the second step, the maximization step, the parameters of the HMM are re-estimated. Hereby, the results from the expectation step and the probabilistic concept of counting occurrences are used the following way,

¹In case $t = T$, $\gamma_T(j)$ is computed by $\sum_{i \in I_S} \xi_{T-1}(i, j)$.

$$\begin{aligned}\bar{\pi}_i &= \mathcal{P}(Z_1 = s_i | \dot{\mathbf{X}}_{\mathbf{T}}) = \gamma_1(i) \\ \bar{a}_{ij} &= \frac{\mathbb{E}(\# \text{transitions from } s_i \text{ to } s_j)}{\mathbb{E}(\# \text{transitions from } s_i)} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)} \\ \bar{b}_{ik} &= \frac{\mathbb{E}(\# \text{visits in } s_i \text{ and observe } o_k)}{\mathbb{E}(\# \text{visits in } s_i)} = \frac{\sum_{t=1}^T \mathbb{1}_{\{o_{k_t}=o_k\}} \gamma_t(i)}{\sum_{t=1}^T \gamma_t(i)}.\end{aligned}$$

Baum [2] shows that the likelihood of the re-estimated parameters $\bar{\lambda} = (\bar{\pi}, \bar{A}, \bar{B})$ is always greater or equal to the likelihood of the previous parameter setting λ . By iterating this procedure, the parameters are improved and thus, provides an applicable training method. However, it only converges against a local minimum, not guaranteeing an optimal solution to the training problem. Nonetheless, it has been shown in practice that in general this local minimum is sufficient and additionally not very sensitive to different initial settings. One possible reason is that the parameter space is highly symmetric due to interchangeability of the hidden states. Therefore, we choose the initial parameters at random. The algorithm terminates if the log-likelihood has converged or a maximum number of iterations is reached.

Baum-Welch algorithm

1. *Initialization*

$\lambda = (\pi, \mathbb{A}, \mathbb{B})$ is chosen randomly

For i in $1, \dots, \text{maxIteration}$:

2. *Expectation step*

$$\begin{aligned}\xi_t(i, j) &= \frac{\tilde{\alpha}_t(i) \cdot a_{ij} \cdot b_{j_{k_{t+1}}} \cdot \tilde{\beta}_{t+1}(j)}{\sum_{i' \in I_S} \sum_{j' \in I_S} \tilde{\alpha}_t(i') \cdot a_{i'j'} \cdot b_{j'_{k_{t+1}}} \cdot \tilde{\beta}_{t+1}(j')} \\ \gamma_t(i) &= \sum_{j \in I_S} \xi_t(i, j)\end{aligned}$$

3. *Maximization step*

$$\begin{aligned}\bar{\pi}_i &= \gamma_1(i) \\ \bar{a}_{ij} &= \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)} \\ \bar{b}_{ik} &= \frac{\sum_{t=1}^T \mathbb{1}_{\{o_{k_t}=o_k\}} \gamma_t(i)}{\sum_{t=1}^T \gamma_t(i)}\end{aligned}$$

4. *Termination condition*

if $\mathcal{P}_{\lambda}(\dot{\mathbf{X}}_{\mathbf{T}}) = \mathcal{P}_{\bar{\lambda}}(\dot{\mathbf{X}}_{\mathbf{T}})$: terminate
else: set $\lambda = \bar{\lambda}$

return λ

2.1.3. HMM Forecasting

This section introduces the theoretic reasoning behind HMM forecasting methods. Given the efficient algorithms of the previous section, this task is derived straight forward.

Additional, while we aim to use the resulting probabilities for probabilistic forecasts, we show possible methods to apply HMM in point forecasting.

The general goal of the probabilistic forecasting problem is to predict with which probabilities future events are occurring. Specifying a fixed future time horizon $H \in \mathbb{N}$ and given the historic observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$, we aim to calculate the probability

$$\mathcal{P}(X_{T+H} = o_k | \dot{\mathbf{X}}_{\mathbf{T}})$$

for all possible observations $k \in I_O$. Due to the HMM properties, these probabilities are calculated by tracing back all possible hidden state sequences probabilities leading up to the forecasting time instance.

First, we break down the initial probability into the following,

$$\begin{aligned} \mathcal{P}(X_{T+H} = o_k | \dot{\mathbf{X}}_{\mathbf{T}}) &= \sum_{j \in I_S} \mathcal{P}(X_{T+H} = o_k | \dot{\mathbf{X}}_{\mathbf{T}}, Z_{T+H} = s_j) \mathcal{P}(Z_{T+H} = s_j | \dot{\mathbf{X}}_{\mathbf{T}}) \\ &= \sum_{j \in I_S} b_{jo_k} \mathcal{P}(Z_{T+H} = s_j | \dot{\mathbf{X}}_{\mathbf{T}}) \\ &= \sum_{j \in I_S} b_{jo_k} \sum_{i \in I_S} \mathcal{P}(Z_{T+H} = s_j | Z_T = s_i) \mathcal{P}(Z_T = s_i | \dot{\mathbf{X}}_{\mathbf{T}}) \end{aligned}$$

From Markov theory, we can identify the probability $\mathcal{P}(Z_{T+H} = s_j | Z_T = s_i)$ as the H -step-transition probability $a_{ij}^{(H)}$, see This quantity is computed using the H -step-transition matrix $\mathbb{A}^{(H)}$ for which the Chapman-Kolmogorov equation shows that

$$\mathbb{A}^{(H)} = \mathbb{A}^H.$$

Further, the second probability term is the already known quantity $\alpha_T(i)$ from the forward algorithm, $\alpha_T(i) = \mathcal{P}(Z_T = s_i | \dot{\mathbf{X}}_{\mathbf{T}})$. Let α_T denote a row vector with elements $\alpha_T(i)$. Then we can formulate equation 2.1 in matrix notation.

$$\begin{aligned} \mathcal{P}(X_{T+H} = o_k | \dot{\mathbf{X}}_{\mathbf{T}}) &= \sum_{j \in I_S} b_{jo_k} \sum_{i \in I_S} a_{ij}^{(H)} \alpha_T(i) \\ &= \sum_{j \in I_S} b_{jo_k} [\alpha_T \mathbb{A}^H]_j \\ &= \alpha_T \mathbb{A}^H \mathbb{B}_{.o_k} \end{aligned} \tag{2.1}$$

Hereby, $[\alpha_T \mathbb{A}^H]_j$ denotes the j -element of the row vector $\alpha_T \mathbb{A}^H$ and $\mathbb{B}_{.o_k}$ is the o_k -column of the observation matrix $\mathbb{B}$. We introduce the M -dimensional row vector θ representing the desired distribution vector over the observation space

$$\theta = [\mathcal{P}(X_{T+H} = o_k | \dot{\mathbf{X}}_{\mathbf{T}})]_{k \in S_O}.$$

Considering equation 2.1.3, the solution for the forecasting problem can then be written in the following compact form,

$$\theta = \tilde{\alpha}_T \cdot \mathbb{A}^H \cdot \mathbb{B}.$$

In this equation, we can observe well the mechanism of the HMM forecasting procedure. The vector α_T is the distribution of the hidden Markov process during the time step T (representing all available information of historical data). If we multiply this distribution by the transition matrix $\mathbb{A}$, we derive the probabilities for the position of the hidden Markov process of the next time step. Repeating this matrix multiplication h -times, we result in the distribution of the hidden Markov process at the time instance $T + H$. In the final step, we multiply the observation matrix $\mathbb{B}$ that results in the desired distribution of the observable process.

Summarizing the forecasting process within an algorithm consists therefore of two steps: First, computing $\tilde{\alpha}_T$ via the adapted forward algorithm and second, calculating θ through matrix multiplication.

HMM Forecasting Algorithm

1. Step:
Calculate $\tilde{\alpha}_T$ via the adapted forward algorithm.
2. Step:
 $\theta = \tilde{\alpha}_T \cdot \mathbb{A}^H \cdot \mathbb{B}$

return θ

The computational complexity of a single isolated forecast is of the order $O(N^2 \cdot (T + H))$. Even though this computational effort is already viable, the forecasting efficiency can be improved by the following thoughts. Given that in the forecasting step the parameters and thus N are fixed, the two influencing factors are the historic window length T and the forecasting horizon H .

First, discussing the historical factor, recent historic events greatly influence the future outcome, while events longer ago do not bring much additional information. Although using all available data maximizes the available information, the question is how much historic data is truly needed to include all essential information and achieve equally good predictions. Depending strongly on the parameters of the specific HMM, the historic window length T can be reduced to a reasonable length accordingly. In the upcoming chapters of this work, we present an exemplary analysis of the correct choice of T . Furthermore in practice, when new forecasts are generated for each new time instances (for example at time instance T and then at $T + 1$) the previous calculated quantities $\tilde{\alpha}_T$ can be reused for the computation of the new $\tilde{\alpha}_{T+1}$, which cuts down the computational time significantly.²

Secondly, turning to the future horizon H , we can assess the quality of the forecast. Generally, the farther ahead in the future the predicting event is, the less precise the forecast it will be. In the HMM setting, this is determined by the characteristics of the transition matrix $\mathbb{A}$. Given that the hidden Markov process fulfills certain conditions³,

²For scientific reproducibility, this idea is not applied in the experiments of this thesis.

³For irreducible, aperiodic and recurrent Markov chains.

the process converges to the stationary distribution $\delta = \delta \cdot \mathbb{A}$. Thus, also the forecast $\tilde{\alpha}_T \cdot \mathbb{A}^H \cdot \mathbb{B}$ converges with increasing H towards a fixed distribution (which is the empirical distribution of the observation process). Implying that with increasing H , the informative value of the forecast decreases and thus, putting an upper bound to the reasonable usage of HMM as a forecasting method.

The predicted probabilities $\mathcal{P}(X_{T+H} = o_k | \dot{\mathbf{X}}_{\mathbf{T}})$ contain all information available through the HMM model, and therefore, the resulting distribution vector θ represents the prediction with the maximum amount of information. While in the present thesis, we apply θ directly as a probabilistic forecast, these probabilities can also be utilized as a basis for other types of forecasts. This can be described by the abstract mapping function f using θ as input,

$$f(\theta).$$

Depending on the application, the observation space and objective, f can take on various forms. Here, we present some suitable use cases in the following table 2.1.3, while Gneiting and Katzfuss [12] thoroughly discuss the theoretical reasoning and properties of statistical functionals of probabilistic forecasts.

Function f	Application
$\arg \max$	point forecast for categorical data
$\mathbb{E}$, median	point forecast for numerical data
$\mathbb{V}$, quantiles, confidence interval	measure of uncertainty of the forecast

Table 1.: Overview of functions and their applications.

2.2. Probabilistic Forecasting

Forecasts aim to predict a future event as accurately as possible. In classic deterministic forecasting, one possible outcome is predicted and the discrepancy between forecast and realization - the prediction error - is measured and in machine learning context also minimized. This approach suggests that the exact result can be predicted with a perfect prediction. However, information about future events is by nature incomplete or not fully available, and thus the events themselves are necessarily subject to uncertainty. So, it can be beneficial to acknowledge the stochastic nature of forecasts and look at the problem from a probabilistic perspective. The goal of probabilistic forecasting is to model the uncertain behavior of the future event. This can be done by approximating its probability distribution or, in a more simple form, predicting confidence intervals or quantiles of the desired distribution. A popular example are weather forecast with rain probabilities (e.g. 80% chance of rain tomorrow). In general, meteorology has been and still is the driving field of applied probabilistic forecasting, see Murphy and Winkler [29], since weather forecasts always considered meteorological events part of probabilistic processes. Nowadays, probabilistic forecasting also finds applications like e.g. in financial and economic risk management [14], prediction of low-probability events such as earthquakes [24] or floods [8].

In this section, we summarize the elementary theory of probabilistic forecasting based on the work of Gneiting and Katzfuss [12], who provide an overarching introduction to probabilistic forecasting. The section is structured into the theory of *prediction spaces*, the theoretic foundation, followed by the two most important concepts, *calibration and sharpness*, and finally concluded with *scoring* rules, the evaluation metrics for probabilistic forecasts.

2.2.1. Prediction Space

The theoretic framework for probabilistic forecasting is build on the idea of the *prediction space*, which was first introduced by Gneiting and Katzfuss [12] and developed further in Strähl and Ziegel [33]. The general problem is to predict the distribution for a random event. For our purpose sufficient, we discuss a simplified version of the prediction space based on two components.

First, we consider a one dimensional, real-valued random variable Y on the probability space $(\Omega, \mathcal{A}, P)$ with a cumulative distribution function (CDF). Y is the future event, we aim to predict, and the probability space denotes the underlying probabilistic process. The second component is the prediction described by the random variable F on the same probability space. It takes a value in the space $\mathcal{F}$ of all continuous CDFs and measurable to the sub σ -algebra $\mathcal{A}' \subseteq \mathcal{A}$, hence $\mathcal{F} = \{\hat{F} \text{ is CDF} : \hat{F} \text{ continuous} \wedge \hat{F} \text{ } \mathcal{A}'\text{-measurable}\}$. Thus, F is a random CDF depending on the information set $\mathcal{A}'$, which can be interpreted as the information available to the predictor like historic data or expertise. We will use the naming 'predictive distribution' for the CDF-valued random quantity F further on. The event Y and prediction F are connected in the sense that the predictor constructs F in such a way with the available information $\mathcal{A}'$ that it represents the unknown distribution

of Y as closely as possible. Formalizing this idea, we introduce the phrase *relative ideal* in the following definition.

Definition 2.1. The predictive distribution F is ideal relative to the information set A' if $F = \mathcal{L}(Y|A')$, where $\mathcal{L}(\cdot)$ denotes the conditional distribution of the random variable.

2.2.2. Calibration and Sharpness

In this section, we discuss beneficial properties of our predictive distribution F . Since in practice, it is not possible to observe the conditional distribution $\mathcal{L}(Y|A')$, we require different forms of assessment for the predictive distribution.

The first main idea is that the future event should occur with the same probabilities as predicted. If we could repeat the same experiment multiple times with a fixed predicted distribution $\hat{F} \in \mathcal{F}$, we would use the empiric distribution of Y and compare it to $\hat{F}$. However in general, a repetition is not possible since for each time instance the available information and thus also F changes. To overcome this problem, the *probability integral transform* (PIT) is introduced ⁴.

Definition 2.2. The random variable $F(Y)$ is called the PIT of the predictive distribution F .

The PIT is a transform of the original random variable in the interval $[0, 1]$ and is the value that the predictive CDF attains at the observed event. Assuming that F is the true distribution of Y , we see that the PIT $F(Y)$ follows a uniform distribution. Furthermore, if we repeat the experiment for a different time instance, the resulting PIT should still be uniformly distributed. This property is generalized by the concept of *calibration* and elaborated farther in the following definition and theorem.

Definition 2.3. Given the random variable Y and the predictive distribution F .

- (i) The predictive distribution F is called *marginally calibrated* if $\mathbb{E}[F(y)] = \mathcal{P}(Y \leq y)$.
- (ii) The predictive distribution F is called *probabilistically calibrated* if its PIT is uniformly distributed, thus $F(Y) \sim U(0, 1)$.

[Difference Marginally Calibrated vs probabilistically calibrated]

Theorem 2.4. If the predictive distribution F is ideal relative to the information set A' , then it is also marginally and probabilistically calibrated.

Proof. Since F is relative ideal to A' , we can describe it as $F(y) = \mathcal{P}(Y \leq y | A')$. It follows the proof for marginal calibration,

$$\mathbb{E}[F(y)] = \mathbb{E}[\mathcal{P}(Y \leq y | A')] = \mathbb{E}[\mathbb{E}[\mathbf{1}_{\{Y \leq y\}} | A']] = \mathbb{E}[\mathbf{1}_{\{Y \leq y\}}] = \mathcal{P}(Y \leq y).$$

⁴For the case of discrete random variables, the PIT is formulated similar to definition 2.2 with technical additions for handling discontinuities.

Hereby, the characterization of the probability function via the expectancy of the indicator function $\mathbb{1}$, and the tower rule was applied.

For the probabilistic calibration, we need the fact that F takes a value in $\mathcal{F}$, which implies a continuous, monotonically increasing, surjective function. Thus, there also exists a monotonic increasing, right inverse function F^{-1} , hence

$$F \circ F^{-1} = id, \quad (2.2)$$

whereby id denotes the identity function. Considering the push-forward measure of P by Y onto $\mathbb{R}$, F is also almost surely strictly monotonic and injective. Hence, the right inverse F^{-1} is also almost surely left inverse,

$$F^{-1} \circ F = id \quad a.s. \quad (2.3)$$

It follows for $s \in [0, 1]$,

$$\begin{aligned} \mathcal{P}(F(Y) \leq s) &= \mathcal{P}(F^{-1}(F(Y)) \leq F^{-1}(s)) \\ &= \mathcal{P}(Y \leq F^{-1}(s)) \\ &= \mathbb{E}[\mathcal{P}(Y \leq F^{-1}(s) | A')] \\ &= \mathbb{E}[F(F^{-1}(s))] \\ &= \mathbb{E}[s] = s. \end{aligned}$$

Respectively, the equations are argued by the monotony of F^{-1} , equation (2.3), the tower rule, the marginal calibration of F and equation (2.2). Thus, we have shown that $F(Y)$ follows a uniform distribution. $\square$

The theorem 2.4 also holds for non-continuous random variables, which is shown in Brockwell [7]. This result provides us with a necessary, but not sufficient, condition to assess whether a prediction is possibly relatively ideal. That is, by checking the PIT for uniform distribution. There are several statistical tests to check uniformity, such as the Kolmogorov-Smirnov test or the Chi-squared test, more practical, however, is assess the PIT graphically. The standard method the histogram of the PIT of a sufficient many PIT samples⁵. In example, each 10% interval (0-10%, 10%-20%,...) should occur in 10% of all cases, which implies that each bar of the PIT histogram should have roughly the same height at 10%. The advantage over statistical numeric test is that this approach provides additional information of the behavior of the prediction. Characteristic errors such as skewness and over-/underdispersion can be easily spotted in the PIT histogram, see exemplary figures 1.

A uniformly distributed PIT, however, does not imply that the prediction F is ideal to the information set A' . Uniformity also applies for predictions which are still calibrated but do not use all of the available information. In mathematical terms, it is the case for a prediction relative ideal to another sub-sigma A^* algebras that is coarser than A' ,

⁵Also, the auto-covariance function of PIT is regularly used to study historic dependencies of forecasts when using time series data.

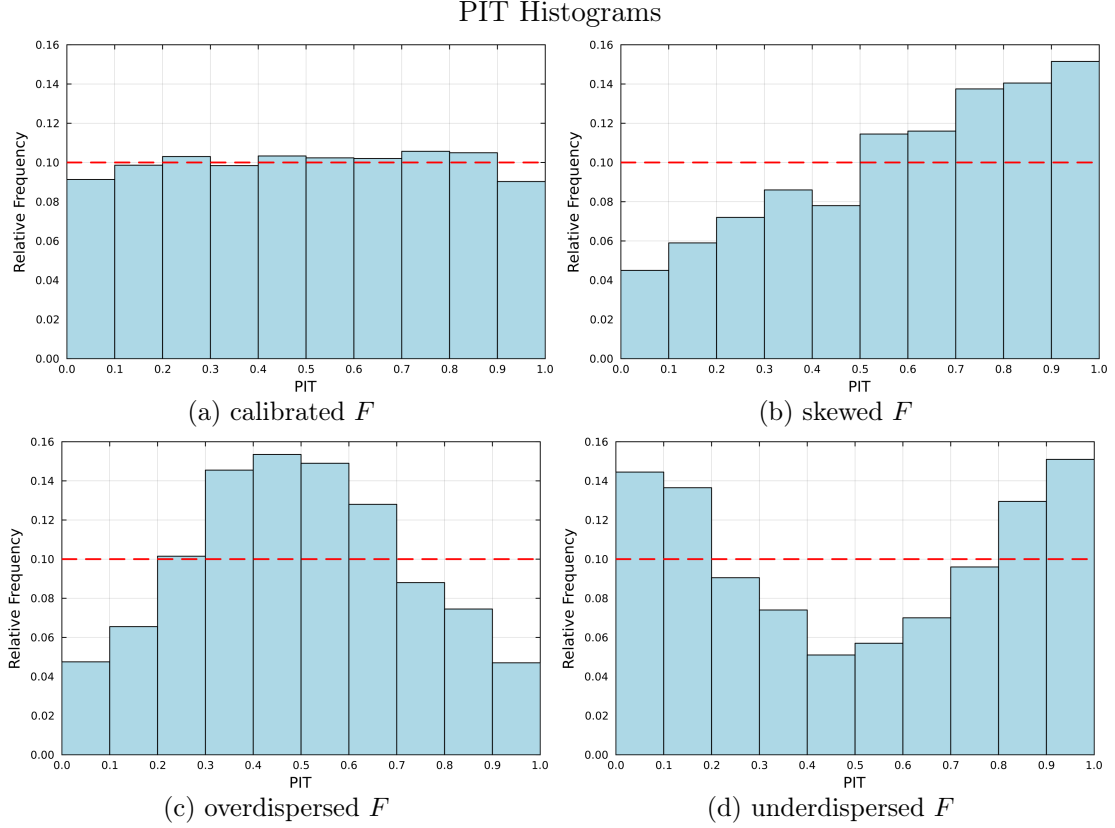


Figure 1.: PIT histogram 1a shows a uniform distributed PIT. In figure 1b the higher intervals occur more often than predicted, while lower intervals occur less often, indicating that the predictive distribution is shifted to the right compared to the underlying distribution. The PIT of figure 1c does not obtain sufficiently often the extreme ends of the distribution implying the predictive distribution to be overdispersed. Whereas in figure 1d, the opposite is the case.

thus $A^* \subset A'$. In example, the unconditional distribution of Y fulfills the condition of a uniformly distributed PIT and is also ideal to the trivial sub-sigma Algebra $A^* = \{\emptyset, \Omega\}$. However, it does not utilize any additional information given by A' . The question is, how can we assess that the maximum of available information was used?

The essential effect of utilizing more information is that uncertainty decreases, thus the predictive distribution becomes generally more concentrated. In probabilistic forecasting, the so-called attribute *sharpness* is used to describe the level of uncertainty. The "sharper" a distribution is, the less uncertain the predictive event is. Given now a set of calibrated predictive distribution, the prediction, which uses the most information, is also the one which maximizes the sharpness. Sharpness can be measured by known metrics like the variance, confidence intervals, or the entropy of a distribution, for example the normal

distribution $N(0, 1)$ is sharper than $N(0, 2)$. Notably, sharpness only takes the predictive distribution into account and does not consider the true realization of the future event. Thus, it evaluates exclusively the predictive distribution.

In summarizing this section, we present the dominating principle of probabilistic forecasting formulated by Gneiting and Katzfuss [12]:

“Probabilistic forecasting has the general goal of maximizing the sharpness of the predictive distributions, subject to calibration.”

2.2.3. Scoring Rules and CRPS

This section discusses how probabilistic forecasts are evaluated numerically. It introduces the concept of *proper scoring rules* and gives a prominent example, the *continuous ranked probability score* (CRPS).

The evaluation metrics of probabilistic forecasts are not as straightforward as those applied in point forecasting. While in point forecasting, the distance between the point prediction and observed value is measured, the aim of the distribution forecast is to match the true underlying distribution, which, however, only "materializes" one observation per instance. Hence, the task is to measure the distance between distribution forecast and observation, and with sufficiently many evaluations, derive comparable results. These measures are called scoring rules, see [30] for a detailed approach.

Definition 2.5. The function $S : \mathcal{F} \times \mathbb{R} \rightarrow \mathbb{R}$ is called a scoring rule.

The scoring rule $S(\hat{F}, y)$ assigns the distribution $\hat{F}$ and the realization y of Y a numerical value. Note that in contrast to the previous section, we consider now a fixed distribution $\hat{F}$, which results from the predictive distribution F .

For the following property definition, another notation of the scoring rule is needed. Given a random variable Y , which follows the distribution $G \in \mathcal{F}$, the expression $S(\hat{F}, G)$ denotes

$$S(\hat{F}, G) = \mathbb{E}_G[S(\hat{F}, Y)].$$

In order to be applicable for the evaluation of probabilistic forecasts, a scoring rule must be *proper*.

Definition 2.6. A scoring rule is *proper* if for all $\hat{F}, G \in \mathcal{F}$

$$S(G, G) \leq S(\hat{F}, G)$$

holds. It is *strictly proper* if $S(G, G) = S(\hat{F}, G)$ implies $\hat{F} = G$.

Thus, a proper scoring rule is designed in such a way that for the true underlying distribution the score is minimized in expectation.

In probabilistic load forecasting, the Continuous Ranked Probability Score (CRPS) has asserted itself in many applications, notably here is the Global Energy Forecasting

Competition (GEFCom2014) summarized in Hong, Pinson, Fan, Zareipour, Troccoli, and Hyndman [21]. Its original form is defined by

$$\mathbf{CRPS}(\hat{F}, y) = \int_{-\infty}^{\infty} (\hat{F}(x) - \mathbb{1}_{\{y < x\}})^2 dx.$$

The elementary idea is to calculate on the left side of the observation the area underneath the predicted distribution and on the right side the area above the predicted distribution, see figure 2. The value is minimized, if the observation lies "central" and the distribution is as sharp as possible. In this way, the CRPS takes the two major concepts of probabilistic forecasting, calibration and sharpness, into account.

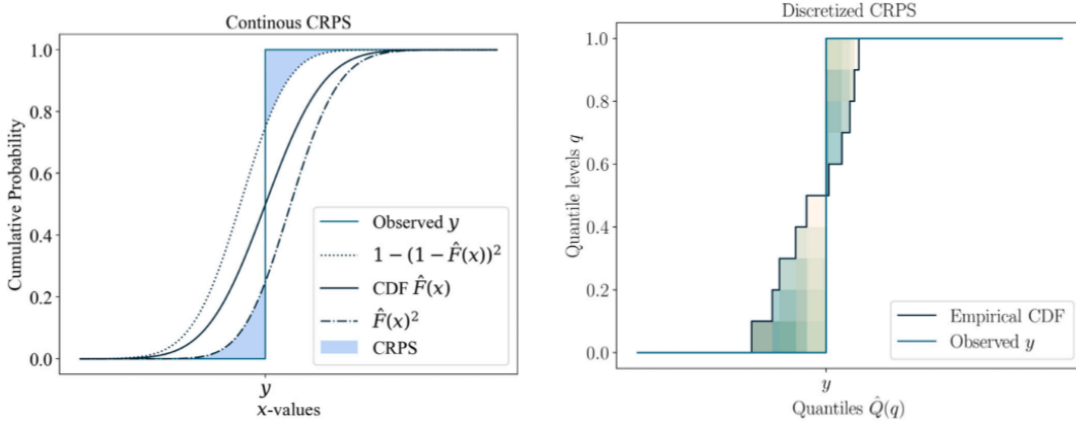


Figure 2.: Graphical example of the continuous and discretized CRPS, reproduced from Botman, Lago, Becker, Vanthournout, and Moor [5]

However, computing the continuous version of the CRPS can be computationally expensive. For this reason, we use a discretized version of the CRPS which is also applied for quantile predictions. This is formulated together with another popular scoring rule, the Pinball score,

$$\mathbf{CRPS}(\hat{F}, y) = \sum_q \mathbf{Pinball}(\hat{Q}(q), y, q)$$

with $\mathbf{Pinball}(\hat{Q}(q), y, q)$:

$$= \begin{cases} q \cdot (y - \hat{Q}(q)), & \text{if } y \geq \hat{Q}(q) \\ (q - 1) \cdot (y - \hat{Q}(q)), & \text{if } y < \hat{Q}(q) \end{cases}$$

where q denotes a quantile and $\hat{Q}(q)$ the predicted value for q of the distribution prediction. For each evaluation, we calculate the quantiles 0, 0.01, 0.02, ... 0.99, 1 from our predicted distribution and compute the CRPS. To achieve comparable results, multiple evaluations are needed and then the mean CRPS is taken.

3. Experiment Design

This chapter provides a detailed explanation on how HMM can be used in the specific use case of forecasting probabilistically the electrical power load of households. The concrete task will be to predict the distribution of the household’s electricity demand in 15 minutes time, only given historical load data. For this reason, we develop an implementation of the HMM forecasting method to predict the electrical load distribution based on the theory presented in the previous chapter. We present the data set of real, historic, electrical power load of households and discuss the applied data processing. The model setup includes a clear structured methodology for building up the HMM forecasting model and a detailed hyperparameter tuning. Further, we conduct a thorough model analysis to study the predictive performance and behavior of the presented HMM model.

3.1. Data

The data source for the experiments presented in this work is the WPuQ dataset collected by Schlemminger, Ohrdes, Schneider, and Knoop [32]. This dataset contains the electrical power load data of 38 single-family houses in northern Germany with the additional information of PV-production, separated heat pump load, weather data, household-specific information and distribution grid measurements. The electrical power was measured over two and a half years during the period of 2018 to 2020 in 10 second intervals. Missing measurements are either labeled as missing value or in cases of missing data gaps smaller than one day linearly interpolated. The authors also provide aggregated data over longer time intervals (1, 15 and 60 minutes) in which the mean power values of the original measurements are taken.

For this thesis, the experiments are conducted on 5 out of the 38 households of the WPuQ. These 5 households were randomly selected with the condition that they do not have PV production and only a minimal number of values are interpolated (less than 1% of the total data). The selection of households are described in table 3.1 and denoted with HH1-HH5 in this paper.

For each of these households, one single time series is taken, namely the total¹ active power load, in 15 minutes resolution. The 15 minutes resolution is chosen because this is the current standard resolution of electricity data in most European and especially Austrian energy systems (like eg. EPEX spot market, smart meter data transmission, etc.). In addition, each of the time series is normalized by dividing the original time

¹Total is in the context of the sum of all three measured phases of the households electric circuit.

Selected Household	WPuQ household ID
HH1	SFH3
HH2	SFH4
HH3	SFH9
HH4	SFH12
HH5	SFH14

Table 2.: Mapping between selected households and WPuQ household IDs

series with its maximum elements. Thus, each value lies in the range of 0 and 1.² We split the data into a training, validation and test datasets. On the training data set, the forecasting models will be trained, the validation data set is used for evaluating different HMM models and finding the optimal hyperparameter setting, and the test set is applied for benchmarking. To avoid seasonal biases, the training data set includes all values of year 2019, which is the first full year with available data. The second fully available year, 2020, is used for validation and testing. Again, to avoid seasonal biases in the evaluation, every odd-numbered month of 2020 (January, March, May,...) is assigned for validation while every even-numbered month of 2020 (February, April, June,...) is assigned to the test data set.

To emphasize the importance of unbiased data sets, Figure 3 demonstrates the time series pattern and differences concerning day times and seasonality. We refer to the paper of Schlemminger, Ohrdes, Schneider, and Knoop [32] for an in-depth analysis of the data characteristics of household’s electrical power demand.

3.2. Model Setup

This section goes through every essential step of the HMM forecasting model applied in the use-case. The target is to predict the distribution of the future electric demand of a household only given historical load data. In order to apply discrete HMMs for this continuous regression problem, a preprocessing and postprocessing of the data is required. Also, we assume training and forecasting as isolated problems and no learning during the forecasting process is allowed. Thus, we only train the model once and use this model for multiple forecasts, although more data may be available in future time steps. The general HMM forecasting pipeline is therefore broken down into preprocessing, training, forecasting, and post-processing.³ In addition and as standard machine learning methodology recommends, we conduct a hyperparameter tuning to obtain the optimal hyperparameter settings whereas most existing literature in HMM forecasting neglected this essential step.

²No negative values are feasible since households with PV production are excluded.

³This methodology is not restricted to load forecasting and can be applied for any other one-dimensional signal.

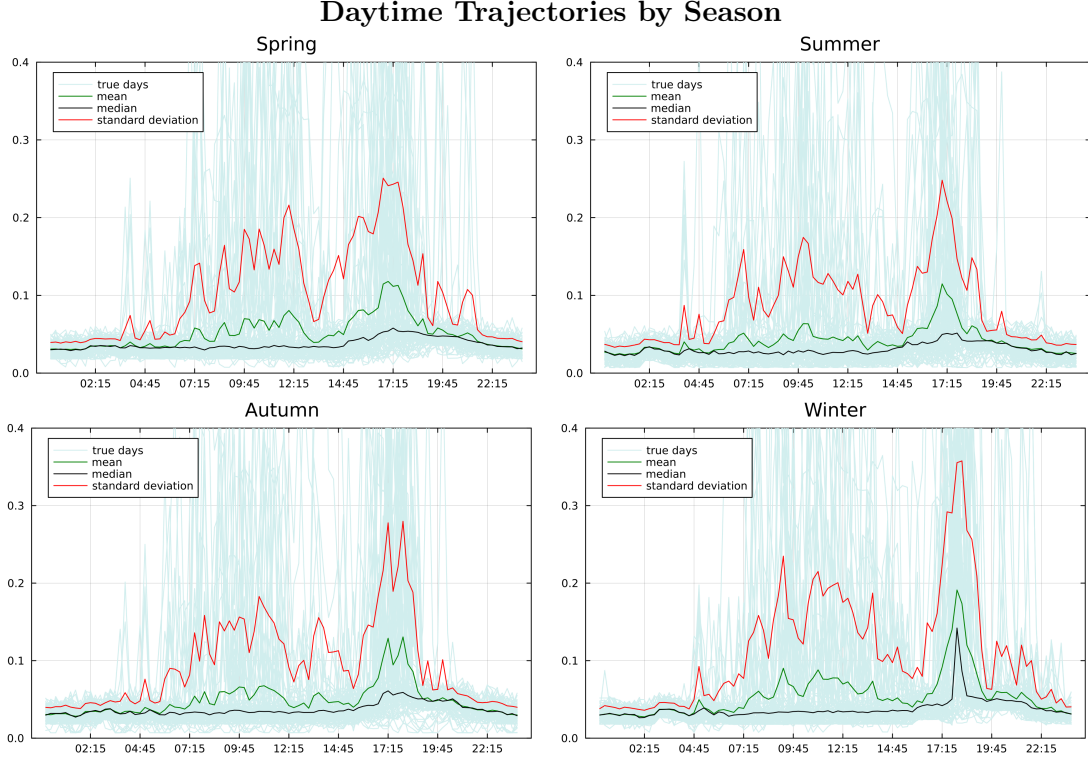


Figure 3.: In these figures for HH1, the daytime trajectories of all days for each season are displayed in gray together with statistics like the mean, median, and mean plus standard deviation. All four seasons show similar load profiles with obvious differences during day and night time. Event hough the profiles of different seasons have a similar shape, the variability of the data is significantly greater during the winter months then during the summer months.

3.2.1. Preprocessing

The historical data is represented by a time series with real valued elements in the electric power unit watts. In preprocessing, this data is discretized from the space of real numbers $\mathbb{R}$ to the discrete observation space O . Thus, there are three variabilities: the number of observations M , the selection of observations in O , and the mapping from $\mathbb{R}$ to O . We assume M as a model hyperparameter, which has to be optimized. The remaining two factors are determined by the classic discretization principle. The original observation space $\mathbb{R}$ is divided into disjunct intervals called bins, with each bin having one representative value. In the discretization process, each real-valued observation is then replaced by the representative value of the corresponding bin in which it is lying in. Since the choice of the intervals and the representative values can have a significant impact on the models performance, we propose two different discretization methods, which both

will be investigated later.

Discretization Type A: Equal-mass bins

The bins are selected so that each bin includes the same number of observations. More precisely, the interval limits of the bins are selected based on the empiric quantiles⁴. As representative value, the median of each bin is chosen. Therefore, high frequented areas of the observation space have a higher resolution than low frequented areas. This discretization method benefits the stochastic design of the HMM model, because the resulting transition and observation probabilities will be balanced due to the balanced occurrences of each discrete observation.

Discretization Type B: Equidistant Bins

The original observation space $\mathbb{R}$ is discretized into bins with equal ranges (like eg. 0-100 Watts, 100-200 Watts, 200-300 Watts, etc.). The representative value of each bin is then the middle point of the bin's intervals. This is the standard discretization method and often desirable from an applied perspective.

3.2.2. HMM Training

The application of HMM to model a one-dimensional signal like electricity demand is designed so that the signal is represented by the observation process $(X_t)_{t \in \{1, \dots, T\}}$ while the hidden process $(Z_t)_{t \in \{1, \dots, T\}}$ is unknown. In our case, we can measure the total amount of electricity demand of a household but do not know the underlying process. This process is influenced by the inhabitants, the electric devices and other factors like weather or season. The advantage of HMM is that this hidden process does not need to be specified. During the training phase, the optimal parameters of the model are generated without deeper knowledge of the process. Via decoding algorithms, the most probable hidden state sequences can be identified and interpreted in a post-training analysis. For example, low energy emitting states can be connected to night time hours or times when nobody is at home. The only condition for the hidden state process is to satisfy the Markov conditions which are quite strong.

The hyperparameters of our HMM model are the type of discretization, the size of the observation space M and the number of hidden states N . We will explain and conduct a detailed hyperparameter analysis later on. During the training phase, the parameters λ of the HMM are trained with the Baum-Welch algorithm given the specific hyper parameter setting and the historic observation sequence $\hat{\mathbf{X}}_{\mathbf{T}}$. The Baum-Welch algorithm is run using the fixed settings of random initial parameters and a maximum number of iterations set to 100 which has been shown to be sufficient. Since the training algorithm relies strongly on the empiric occurrences of the observations, in the selection of the training data data-specific bias should be avoided. In the case of household's electric power load,

⁴For example, when the number of discrete observations is fixed to N , the n 'th bin has as left interval limit the $(n-1)/N$ 'th empiric quantile and as the right limit the n/N 'th empiric quantile.

it is known that there are strong daytime and seasonal differences. Therefore, only data including full days and years should be included.

3.2.3. HMM Forecasting Algorithm

Given a trained HMM with parameters λ , we apply the probabilistic forecasting algorithm and calculated the desired distribution θ as explained in 2.1.3. The two relevant parameters for forecasting are the historic window length T and the future horizon H . For the first parameter T we will use a fixed length and conduct a sensitivity analysis later to verify the choice. The future horizon H is set to 1, thus, it will be a one-step-ahead prediction. Even though it is the most simple forecast, we can derive a lot of information about the quality of the forecasting model because predictions further ahead build upon the same information as the one-step-ahead prediction.

3.2.4. Postprocessing

The resulting prediction from the HMM forecast is the discrete distribution vector θ over the discretized observation space O , however, the original problem requested a distribution forecast in the continuous space $\mathbb{R}$. For this reason, the discrete distribution is re-converted to a continuous CDF with the following method. Recalling that the distribution vector θ represents the probabilities of the corresponding bins, we presume that values inside each bin have equal densities. Assuming the bins are sorted ascending, the probability of X_{T+H} being smaller than the right interval limit of bin k is the sum of the probabilities of all bins l lower and equal to bin k ,

$$\mathcal{P}(X_{T+H} \leq \text{right interval limit of bin } k) = \sum_{l=1}^k \theta_l.$$

This leads to a piecewise-linear CDF, where the above defined supporting points are connected by straight lines as shown exemplary in graph 4.

3.2.5. Hyperparameter Tuning

The general goal of the hyperparameter tuning is to find the best HMM forecasting models for one-step-ahead predictions. For this reason, not only the optimal parameter settings are trained, but also the right choice of hyperparameter has to be studied. There are four different hyperparameters that result from the methodology: the discretization method, the number of hidden states N , the number of observations M , and the historic window length for forecasting T . For each discretization method, the optimal settings of N and M are elaborated with a hyperparameter analysis, while the correct choice of T is verified afterwards with a sensitivity analysis. All experiments are conducted for each household and discretization type independently, resulting in two final HMM forecasting models per household.

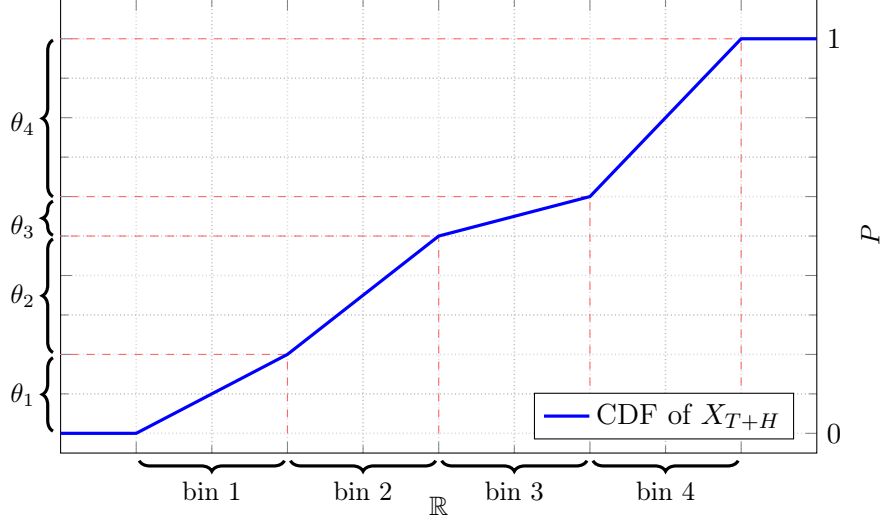


Figure 4.: Continuation of the distribution vector θ with four exemplary bins.

Optimization of N and M

For the hyperparameter tuning, we perform a grid search to find the optimal settings of the number of hidden states N and observations M for each household and discretization type A and B . For each hyperparameter setting, the optimal parameters of the HMM model are trained via the Baum-Welch algorithm on the test data set of the household. The models are then evaluated using the validation data set and computing the mean-CRPS of the one-step-ahead prediction for each validation time instance. Hereby, we fix the historic window length T to be 100 which is a length of slightly more than one day considering the data time resolution of 15 minutes. The tested values for the grid search of N and M additional to the other fixed hyperparameters are summarized in table 3. All combinations of these hyperparameter settings are evaluated resulting in the training and testing of 36 models per household and discretization type.

Hyperparameter	Values
N	10, 20, 30, 40, 50, 60, 70, 80, 100
M	25, 50, 100, 200
H	100

Table 3.: Overview of hyperparameter settings used in the hyperparameter analysis.

Sensitivity Analysis for T

In the sensitivity analysis, we investigate the effect of change in the historic window length T on the models forecasting performance. Hereby, we aim to verify the choice of T of the

hyperparameter analysis, but also to get insights of the HMM forecasting behavior. Since the historic window length T is only relevant during the prediction step and not during training, the models do not have to be retrained. Building up onto the previous results, we use the best performing models for each household and discretization type, to which we will refer as *optimal models* for the rest of the paper. With variable values of T , the optimal models are evaluated once again by computing the one-step-ahead prediction for each time instance of the validation data set and calculating the mean-CRPS. The tested values are described in table 4 and the result is analyzed visually.

Hyperparameter	Values
N	fixed to optimal
M	fixed to optimal
H	1, 2, 3, 5, 10, 25, 50, 100, 200

Table 4.: Overview of hyperparameter settings used in the sensitivity analysis.

3.3. Model Analysis

After the optimal models are determined, we investigate the models predictive performance and power in order to gain more understanding of the model’s characteristics. First, the general performance of each optimal is evaluated, before we investigate the predicted distributions in detail. For this purpose, the calibration of the models are examined, and then the behavior of the model for multi-step-ahead predictions is studied. All experiments here are conducted on the test data set of each household and for each optimal model separately.

3.3.1. Evaluation

Similar to hyperparameter tuning, the models are evaluated using the discrete mean-CRPS. Thus, for each optimal model, the one-step-ahead distribution is predicted for each time instance of the corresponding household’s test data set and evaluated. The result will be for each household 2 different error measures on which we can deduct a good comparison of the discretization types.

3.3.2. Calibration via PIT Histograms

While the mean-CRPS already considers calibration, it is also influenced by the sharpness of the predicted distribution. To gain more insight into the predictive power of the models, we generated the PIT histograms for the optimal models, as presented in section 2.2.2. Thus, for each time instance of the test data set, the PIT value is calculated between the predicted distribution and the true realization, and then the histogram of these PIT values is generated. We structure the PIT histogram into 10 sections displaying the frequencies for each 10% interval (0-10%, 10-20%, etc.). This provides a visual judgment

of the model’s calibration which is sufficient for our purpose. In literature, there exist some additional numeric and theoretic in-depth analytic methods like the reliability index [13] or calibration vs sharpness decompositions of the CRPS [1].

3.3.3. Convergence for Increasing Forecasting Horizon

The last study of our model investigates the model behavior for a forecasting horizon larger than one, as applied until now. We are not interested in the numerical performance of the models; however, we want to know how far into the future our HMM models can be reasonably used. For this reason, we investigate the convergence of the predicted distribution θ with increasing forecasting horizon H . Convergence means that the predicted distribution does not change after a certain point in the future. To illustrate the idea of convergence, figure 5 shows a typical situation of a multi-step-ahead distributional forecast. Generally speaking, the convergence of a distributional forecast can be meaningful in certain applications. However, knowing that there are daytime patterns within the power load data, the convergence of the forecast only shows the limitation of the HMM model which cannot model long-term patterns.

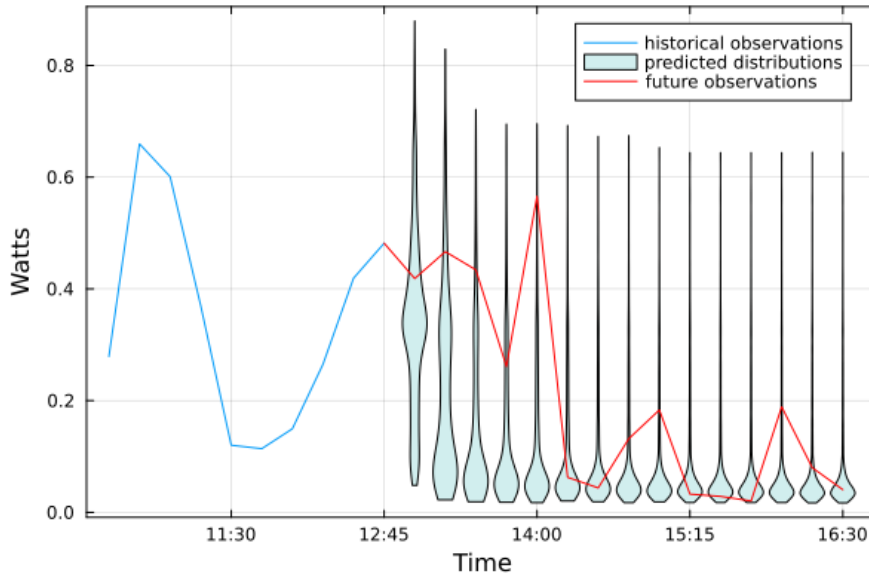


Figure 5.: Example of a typical situation of a multi-step-ahead distribution forecast. Given the historic observation, a distribution forecast is computed for each time instance in the future illustrated by the multiple violin plots. Additionally, the true realization of the future time instances were added to the plot afterwards. Examining the shape of the single distributions, the convergence is visible. From time instance 14:00 onward, the violins do not change significantly and from 15:15 onward, they are visually not differentiable.

Formulating the idea of convergence in mathematical terms, the stochastic vector θ

lies on a hyperplane in the N -dimensional real space. We denote θ_h as the predicted distribution h steps into the future, for $h \in \{1, \dots, H\}$. The time step h' is defined as the *convergence point* if for all h greater than equal to h' , θ_h lies inside the epsilon-neighborhood around the empiric distribution of the observation process,

$$\|\theta_h - (\delta \cdot \mathbb{B})\|_1 < \epsilon \quad \text{for all } h \geq h'.$$

Here, δ denotes the stationary distribution of $\mathbb{A}$, and thus, $(\delta \cdot \mathbb{B})$ is the empiric distribution. The L^1 metric is chosen, so that the choice of epsilon defines the absolute difference between the stochastic vectors. For example with $\epsilon = 0.01$, convergence is achieved when the predicted distribution vectors differ only up to 1% in total. We experiment with two different epsilon settings, first with $\epsilon = 0.1$ for a rough convergence of the distributional shape and $\epsilon = 0.01$ for a detailed convergence.

4. Results Hyperparameter Tuning

4.1. Optimization of N and M

For each household and both discretization types, 36 HMM models with different hyperparameter settings were trained and evaluated. Figures 6-10 visualize the results of the hyperparameter tuning for each household. In each figure, a plot for each discretization type shows the mean-CRPS value in dependence of different settings of hidden states numbers, while models with equal number of observation are visualized by equal color. For each plot, the hyperparameter setting inducing the lowest mean-CRPS value is chosen for the optimal model. In appendix A, the reader finds a complete table with the exact numerical results of the hyperparameter tuning.

Household	N	M	CRPS	Household	N	M	CRPS
HH1	40	100	0.8232	HH1	80	100	0.8259
HH2	30	100	0.6846	HH2	40	100	0.6874
HH3	50	50	1.3658	HH3	50	100	1.3587
HH4	60	100	0.9505	HH4	50	100	0.9157
HH5	40	100	0.8219	HH5	80	100	0.8213

(a) Discretization Type A
(b) Discretization Type B

Table 5.: The hyperparameter setting of the optimal models for each household

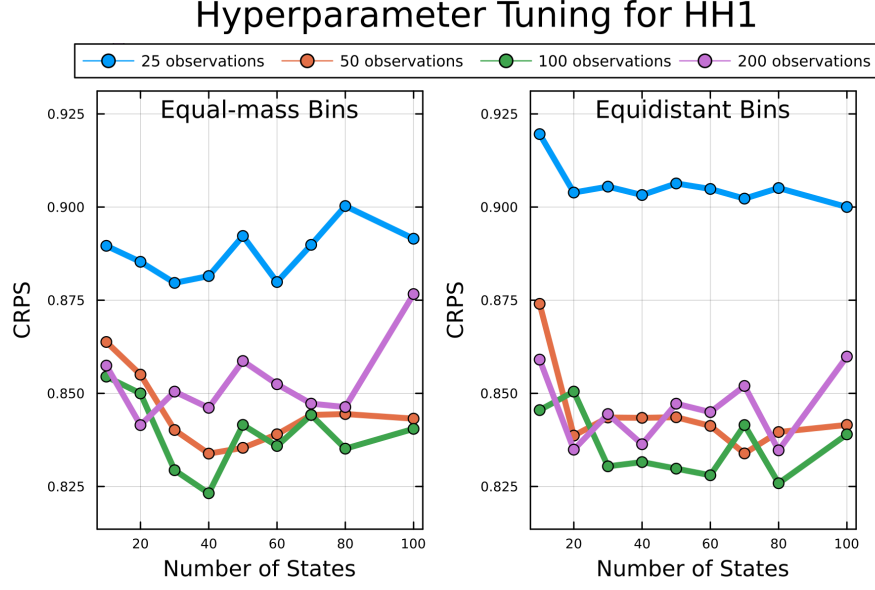


Figure 6.: The optimal models are chosen by the minimal mean-CRPS value, which is for discretization type A the model with 100 observations and 40 hidden states and for discretization type B the model with 100 observations and 80 hidden states.

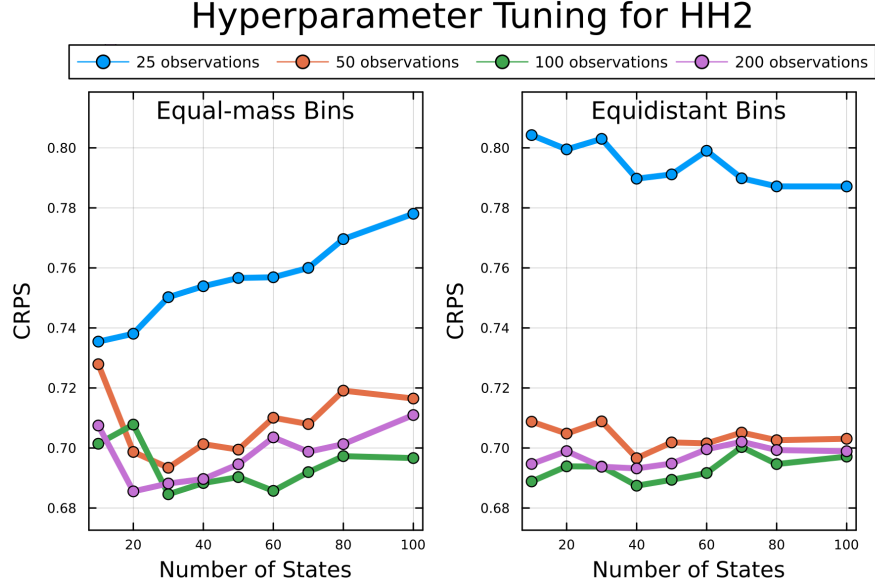


Figure 7.: For discretization type A, two hidden states settings ($N = 30$ and $N = 60$) achieve similar optimal results for models with 100 observations. The development of the mean-CRPS for increasing number of states have a similar trajectory between models with equal number of observation, while the setting with 100 observation achieves the optimal result at 40 hidden states.

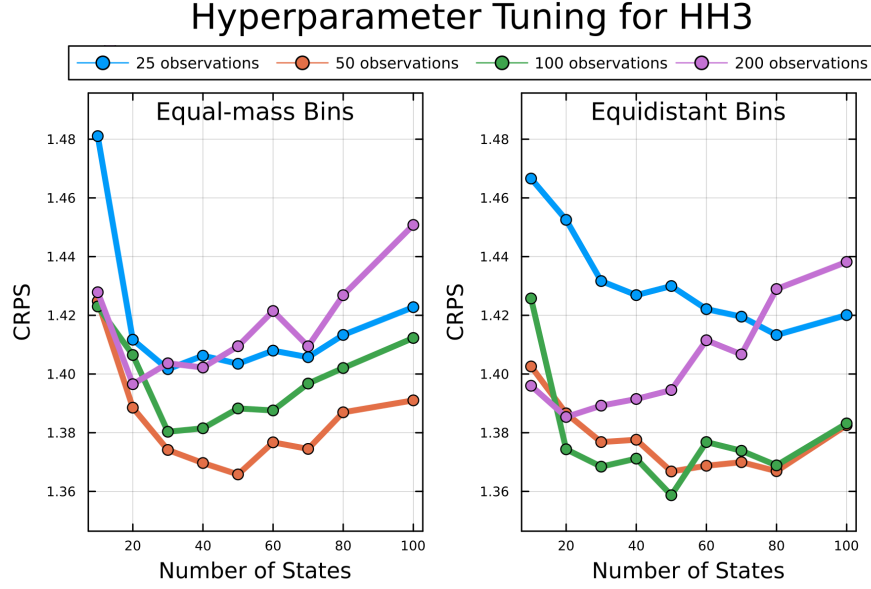


Figure 8.: HH3 has the highest mean-CRPS overall. Also it is the only household, where a model with 50 observation outperforms the rest, in the case of equal-mass bins with $N = 50$. For the other discretization type, the optimal model is set by $N = 50$ and $M = 100$

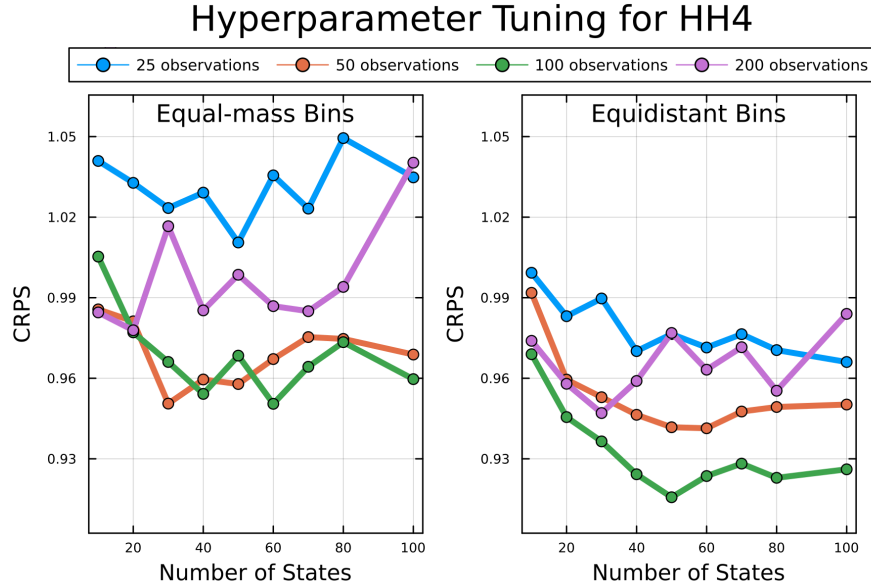


Figure 9.: For discretization type A, two different models achieve a similar optimal mean-CRPS value, with the model for $N = 60$ and $M = 100$ is surpasses by a small margin. For discretization type B, models with 100 observation are consistently better and optimal at $N = 50$.

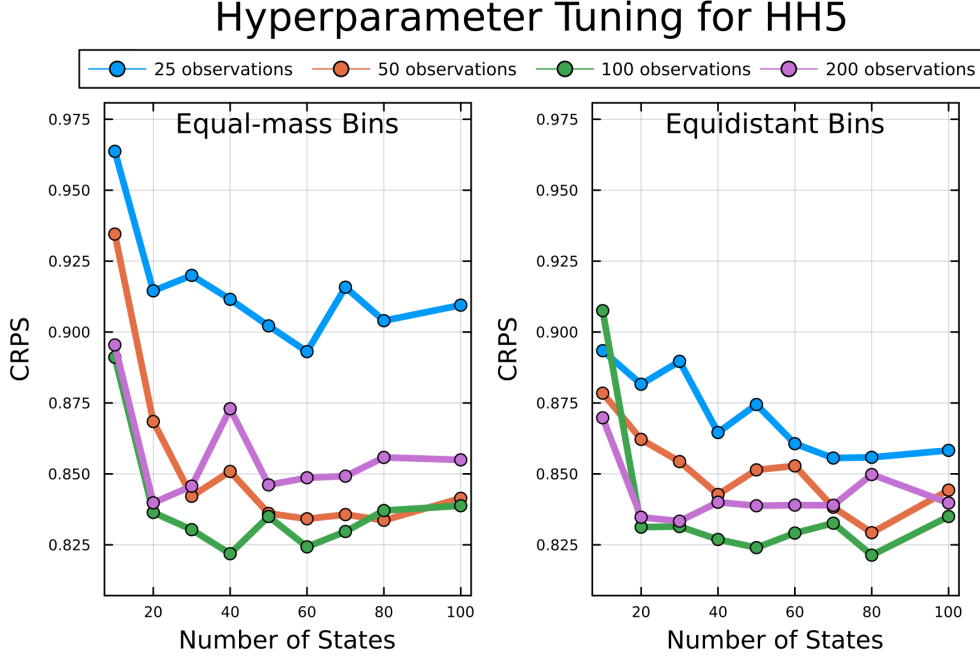


Figure 10.: HH5 shows similarities to HH1 in the aspect of optimal hyperparameter setting and mean-CRPS value for both discretization types (N=40, M=100 for type A; N=80, M=100 for type B).

A summary of the numeric results is given by table 5, which highlights the hyperparameter settings of the optimal models with their corresponding mean-CRPS values. From the hyperparameter analysis figures, the following conclusions are drawn.

First, the number of observations have a significant impact on the models performance. The choice of discretizing the observation space in only 25 observations leads in almost every setting to the worst results, but also a high resolution of 200 observations influences the performance of the models badly. The number of observations of 50 and 100 achieves the most consistent results with 100 observations being the best setup for all households, except for HH3 with discretization type A.

Further, the optimal number of hidden states differs for each household. However, a major result of this analysis is that models of discretization type A reach their optimal setting with lower a lower number of hidden states than models of discretization type B. This effect is explained due to the design of discretization. The discretization of equal-mass bins favors the stochastic characteristic of the HMM models because it leads to a balanced weighting of all observation and thus the underlying hidden states.

4.2. Sensitivity Analysis for T

The sensitivity analysis investigates the convergence of the prediction performance conditioned by increasing the historic window length T . The experiments were conducted for each household and discretization type with the optimal models hyperparameter setting summarized in 5. The results are visualized by figures, however, since they do not differ between the household significantly only one representative figure 11 for HH1 is presented here, while the corresponding figures for HH2-HH5 are in the appendix B.

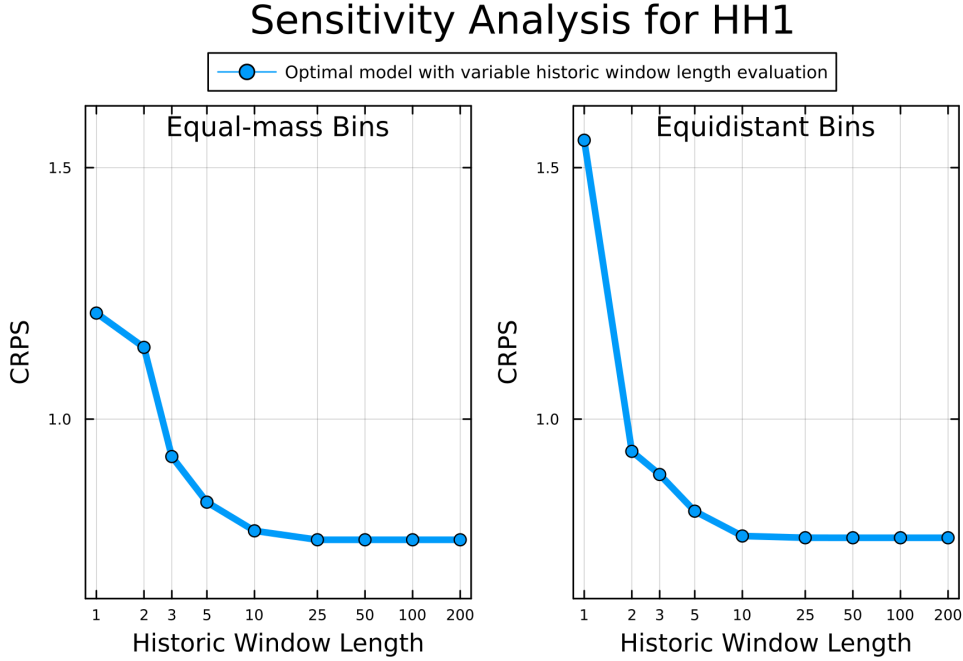


Figure 11.: Representative figure of the result of the sensitivity analysis for HH1.

The results of the sensitivity analysis support the hypothesis that with increasing the historic window length and thus increasing available information, the performance of the forecasts improves. All households show a similar pattern of converging mean-CRPS to a lower limit. This limit is reached in most cases at $T = 10$, for HH2 already at $T = 5$ and only in the case of HH1 with discretization in equal-mass bins at $T = 25$. There are no significant differences between the discretization types, even though they differ greatly in the number of hidden state, like eg. in HH1 and HH5.

In addition, it verifies that the choice of T during the hyperparameter analysis was well taken. For further experiments in the future, the historic window length can be chosen even lower to improve computational evaluation times.

Lastly, considering the converging point to be around 10, it can be said that only historic data up to two and a half hours ago adds beneficial information to the HMM forecasting model. That also implies that the model does not detect recognizable behavior and patterns before this point, inferring that HMM forecasting models should only be applied

for short-time pattern forecasting.

5. Results Model Analysis

This chapter presents the results of the model analysis and discusses their implications to better understand the model’s characteristics. For this, the optimal models resulting from the hyperparameter analysis in section 4.1 are tested on the test data set of their corresponding household. First, we evaluate the optimal model’s performance, than we investigate the calibration of the predicted distribution and in the end, analyze the models behavior for multi-step-ahead predictions.

5.1. Evaluation

Household	N	M	CRPS	Household	N	M	CRPS
HH1	40	100	0.802	HH1	80	100	0.812
HH2	30	100	0.720	HH2	40	100	0.723
HH3	50	50	1.363	HH3	50	100	1.357
HH4	60	100	0.956	HH4	50	100	0.901
HH5	40	100	0.819	HH5	80	100	0.831

(a) Discretization Type A

(b) Discretization Type B

Table 6.: The hyperparameter setting of the optimal models for each household

Table 6 shows the evaluation results of the optimal models together with their hyperparameter setting. The historical window length T was fixed to be 30, which is sufficient as shown in section 4.2.

Comparing the performance of the discretization methods, we observe for all households except HH4 that both discretization methods achieve similar mean-CRPS values not varying greater than 2%. HH3 stands out with a much higher mean-CRPS than the other households, which can be caused by either a greater volatility of the electricity demand or a conceptual drift in energy consumption for the test data compared to the training data. Also interestingly, in comparing the models for each household separately, we observe that the model with less hidden states performs better than the second model of the household.

5.2. Calibration via PIT Histograms

The PIT histograms are shown in figures 12 to 16. The underlying pit values are calculated based upon the same settings and distribution forecasts as in the previous section.

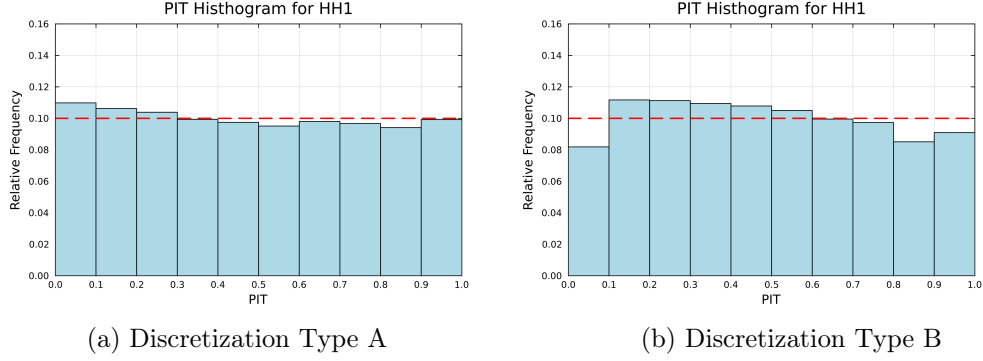


Figure 12.

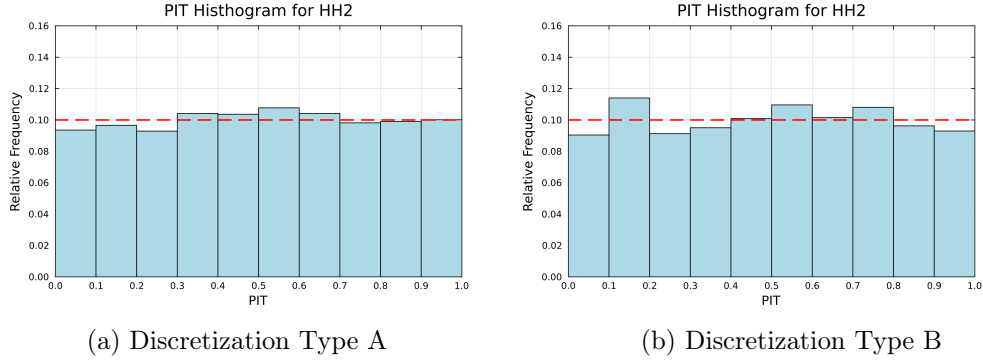


Figure 13.

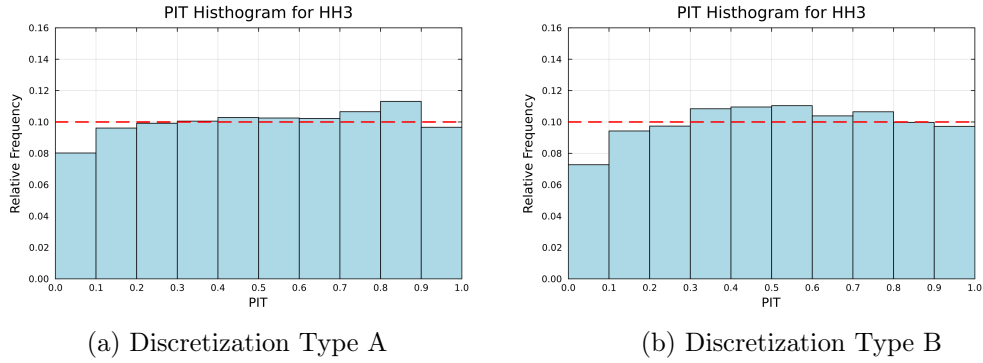


Figure 14.

From these PIT histograms, it is visible that models with discretization type *B* have a worse calibration than models with discretization type *A* throughout all households. This effect can be seen especially strong for HH4 and HH5. While the HH4 model

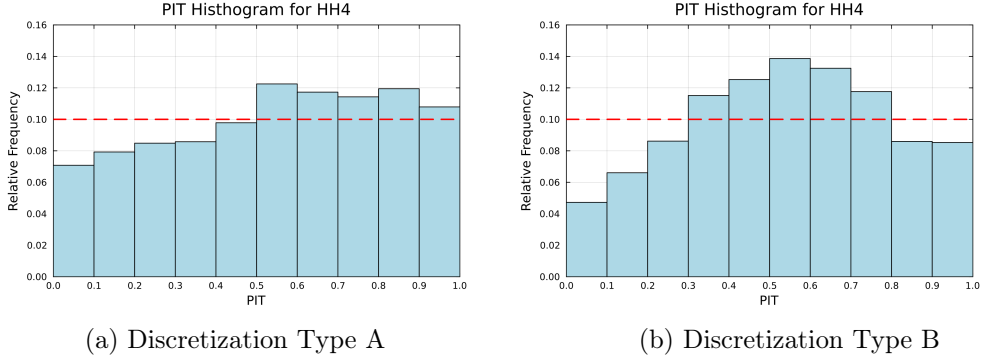


Figure 15.

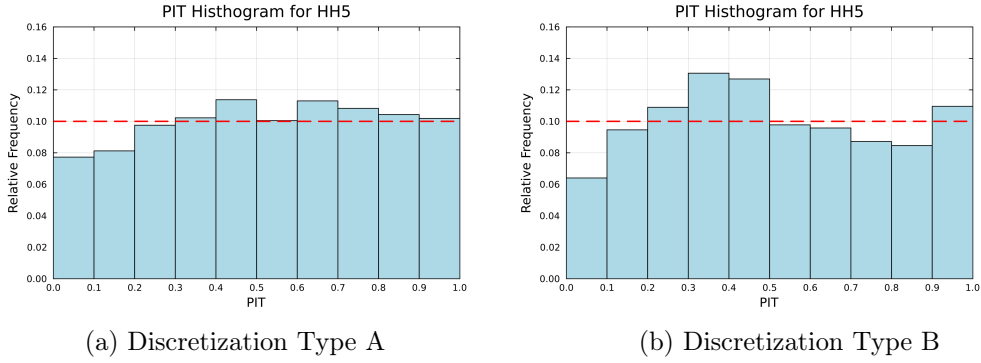


Figure 16.

with equal-mass bins shows a slight skewness towards higher quantiles, the forecast of the HH4 model with equidistant bins is strongly overdispersed. This implies that the reliability of the prediction increases if the discretization with equal-mass bins is carried out. Considering the similar values of the mean-CRPS in the households for different discretization methods, we can assume that discretization type *B* models compensate the lack of calibration with a greater sharpness. In conclusion, we can state that the choice of discretization has an impact on the calibration of the models forecasts. Following the main principle of probabilistic forecasting (see section 2.2.2), models with discretization into equal-mass bins should be favored.

5.3. Convergence for Increasing Forecasting Horizon

The next graphics display the histograms for all convergence points evaluated over every validation data instance and each optimal model. The figures 17-21 show the convergence points histograms for a tolerance of $\epsilon = 0.1$ and figures Trivially, the convergence to the 10% epsilon-neighborhood is reached earlier than in the 1% epsilon-neighborhood. Referring to the rougher convergence, a lot of forecasts do not significantly change its shape around 5 steps into the future, most forecasts some when before 10 steps, and only

a few forecasts do change significantly their shape only later than 10 (like eg. in HH1.). With a finer convergence of $\epsilon = 0.01$, these boundaries shift to the right. In this case, there are also great differences between the household. Especially HH1 and HH5 require more steps to reach their stationary distributions.

These converging points serve as an absolute limit for T for these HMM forecasting models to be applied in a reasonable way, since they converge to the empirical distribution and do not utilize beneficial information anymore.

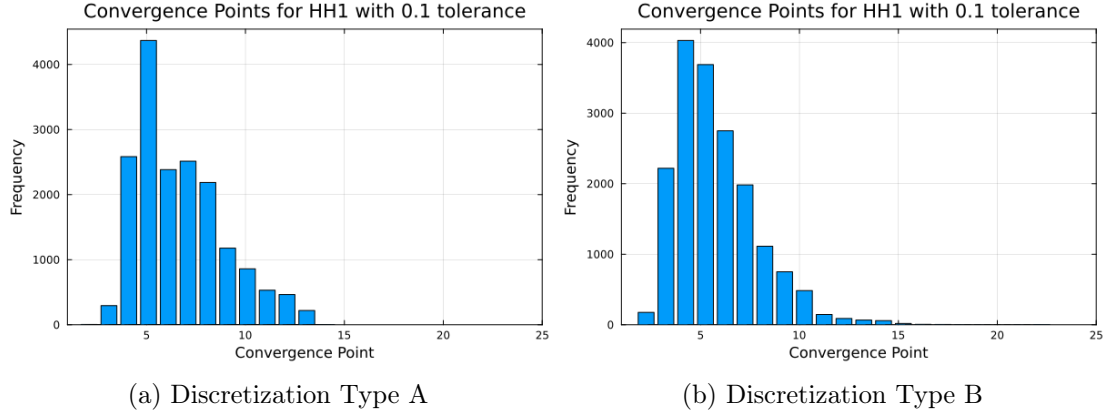


Figure 17.

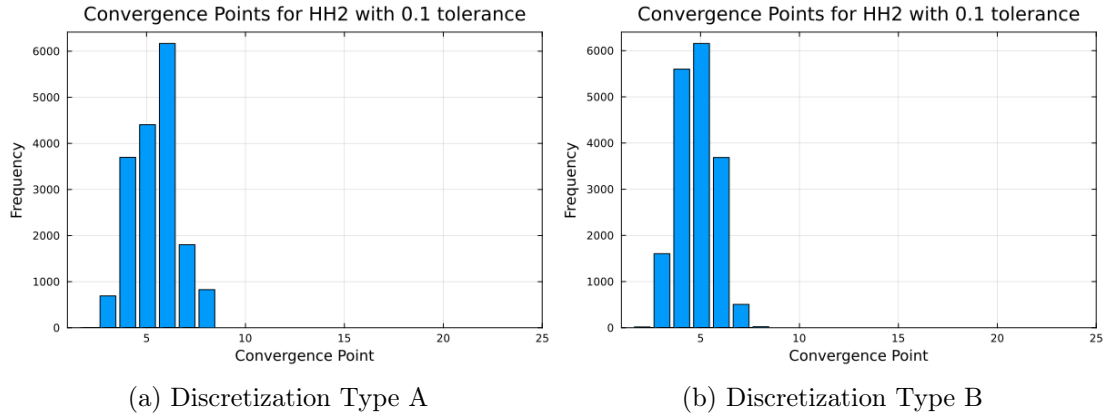


Figure 18.

5. Results Model Analysis

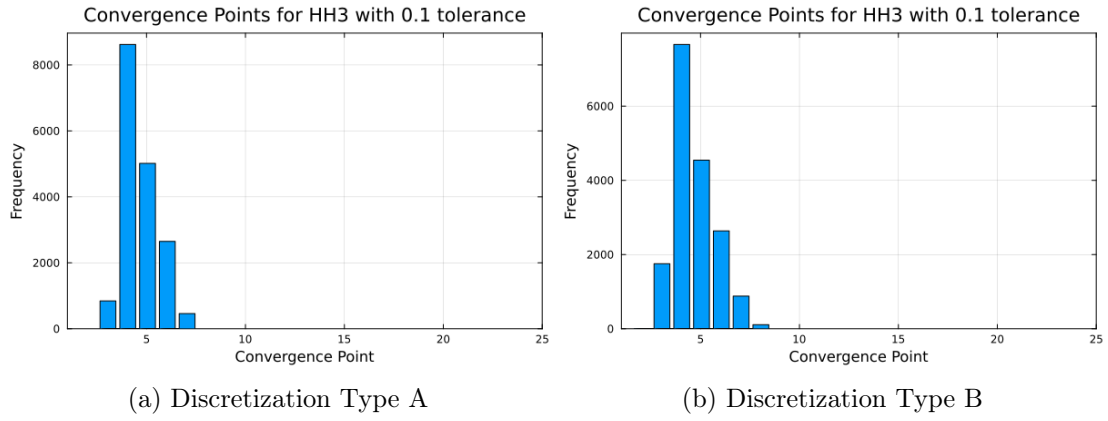


Figure 19.

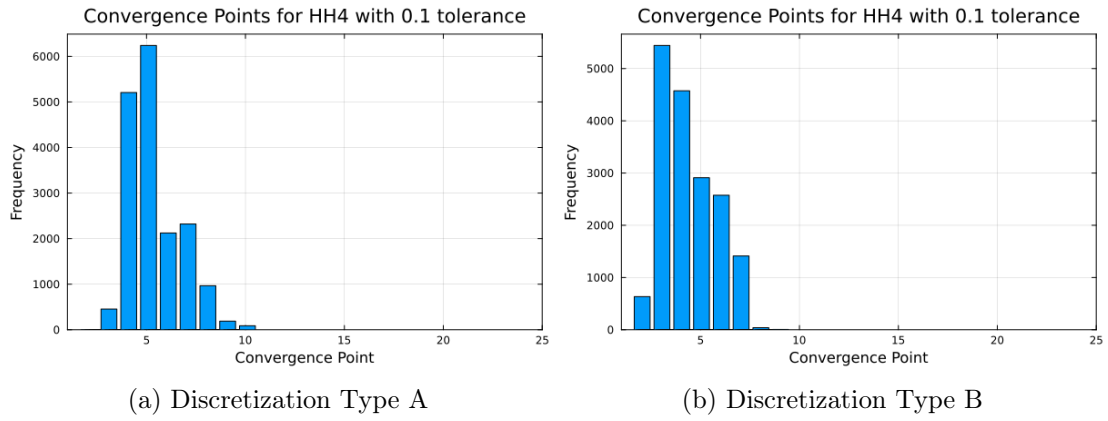


Figure 20.

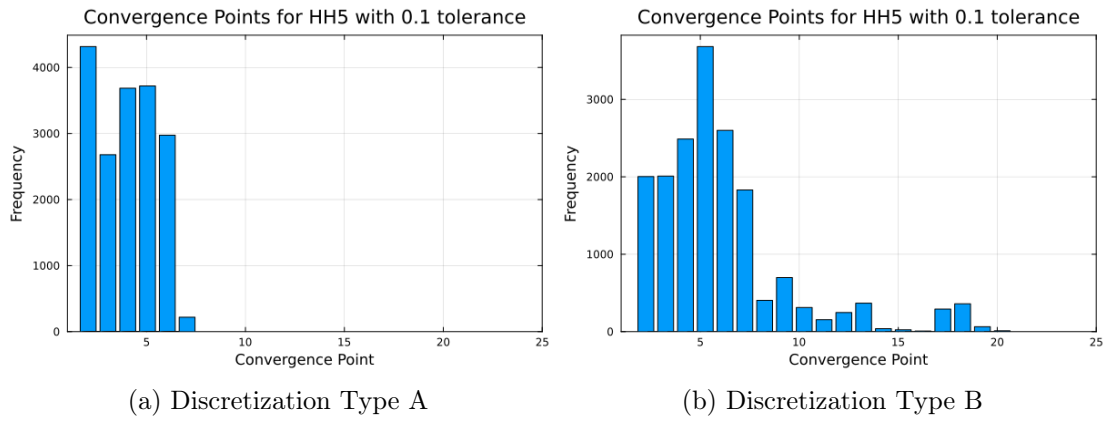


Figure 21.

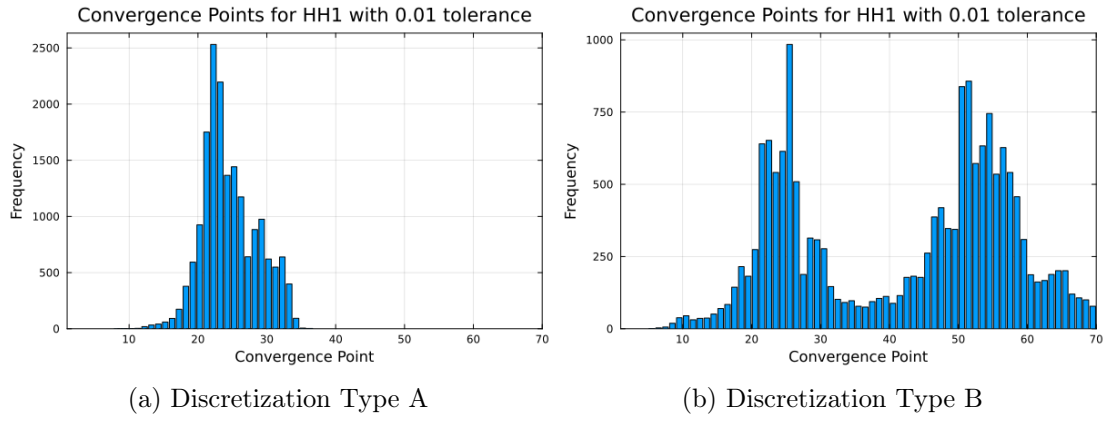


Figure 22.

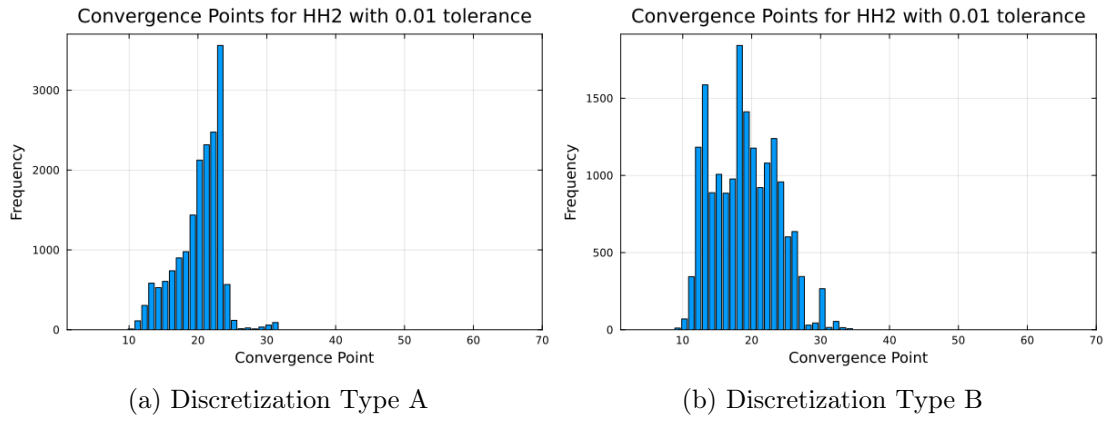


Figure 23.

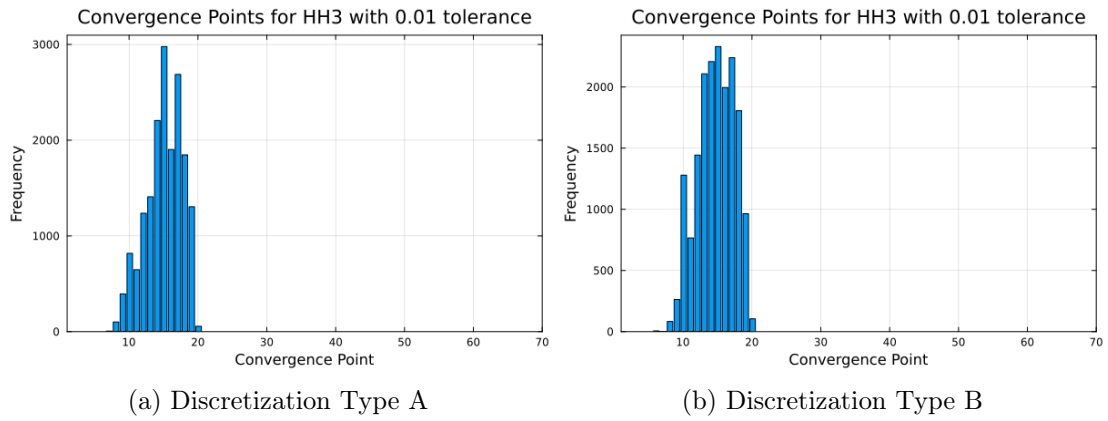


Figure 24.

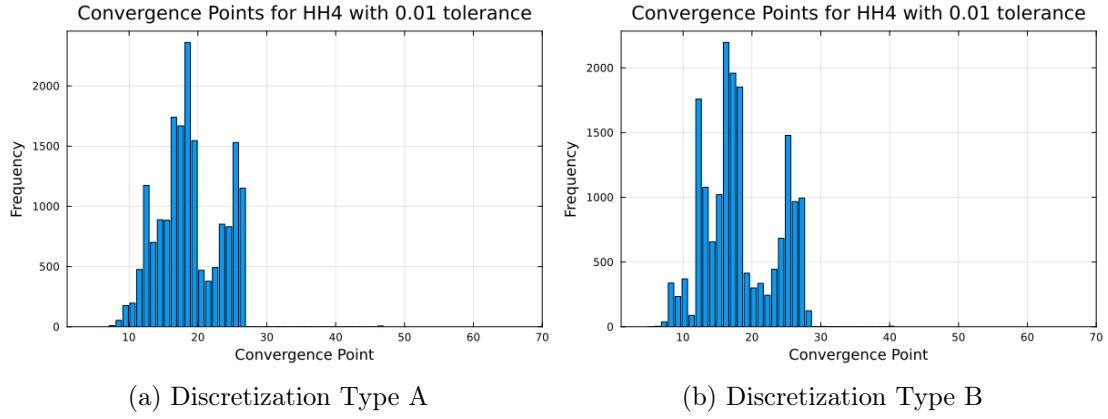


Figure 25.

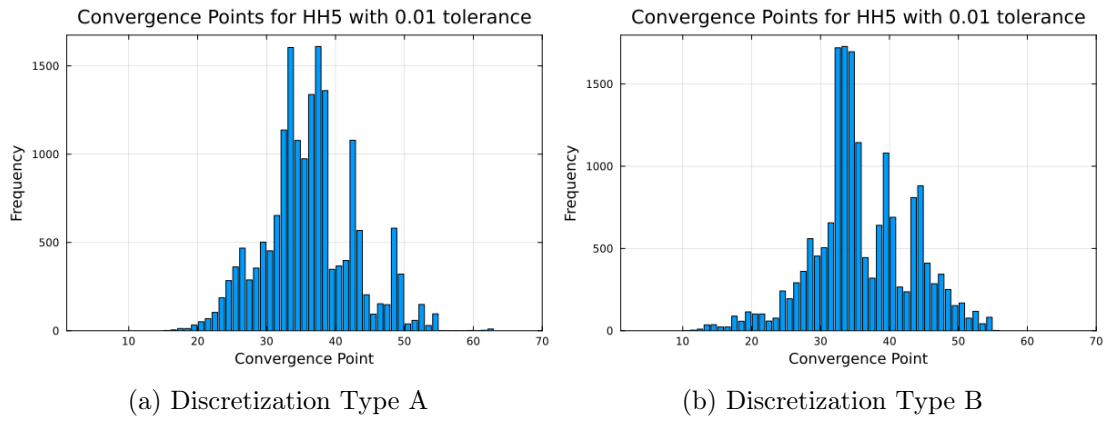


Figure 26.

6. Benchmarks

This chapter addresses the final open question: How does the HMM model perform compared to other state-of-the-art probabilistic short-term power load forecasting methods for households?

The field of load forecasting is diverse, and the consensus of literature is that there does not exist one ultimate model, but rather depends on the specific application and target. A good overview for the different load forecasting methods is given by Haben, Voß, and Holderbaum [17]. Key differences in load forecasting models are namely the forecasting horizon, the voltage level and the application. Our target here is to benchmark with models as similar applicable as the presented HMM forecasting method. Namely, the benchmark models have to be probabilistic, short-term, low-voltage load forecasting model and relatively easy implementable, meaning that both the theoretic and computational complexity of the model is limited.

First, we will state clearly our benchmark framework. Then, we present the benchmark models including two heuristic baselines, one statistic method and one advanced machine learning model. Afterwards follow results and discussion.

6.1. Benchmark Setup

The setup for the benchmarks is alike the experiments conducted in the previous chapters. The general goal is to predict the distribution¹ for the electric power load one-step-ahead. The available information is restricted to historical load data and temporal information, which can be utilized for each benchmark model differently. The historical load data is generally represented by

$$(o_t)_{t=1,\dots,T}$$

for some model specific historic window length $T \in \mathbb{N}$. The available temporal information are daytime or month, which is commonly used to model daytime patterns and seasonality. For this reason, we introduce the two functions *time(.)* and *month(.)*, which define a mapping from the observation's daytime and month of the year onto the interval $[0, 2\pi]$ respectively. This is a common machine learning technique to enable cyclical encoding via $\sin(.)$ and $\cos(.)$, see Bishop [4].

The benchmarks are conducted on the households electrical load data discussed in section 3.1. Thus, the models are trained on the trainings data set and evaluated over the test data set for each household separately. The validation data set can be utilized for model

¹Or a distribution approximation with quantiles.

specific validation tasks. The evaluation is obtained in the same way as in section 3.3.1. For each instance of the test data set, the 99 quantiles for 0.01, 0.02,..., 0.99 are calculated for the one-step-ahead distribution forecast and evaluated with the discrete CRPS. For each model and household, one mean-CRPS value is separately computed. Further, we then compare the results of the benchmark models with the optimal HMM models by their relative improvement to the baseline.

6.2. Benchmark Models

In this thesis, four benchmark models for probabilistic, short-term load forecasting are investigated. Similar models were studied in Botman, Lago, Becker, Vanthournout, and Moor [5], while the reader is referred to Hong and Fan [19] for a detailed discussion of different probabilistic load forecasting methods.

6.2.1. Persistence Model

The most forward baseline method in point forecasting is the so-called *naive* or *persistence* model. The idea is that the predicted value of the future is equal to the value of the presence. To apply this concept as a probabilistic forecast, we use the deterministic naive forecast as a basis and add to this forecast a distribution. This distribution is constructed by applying the naive point forecast on the training set, taking all the error terms and computing the empiric error distributions for each quarter hour of the day separately. The resulting predicted distribution of the persistence model consists of the previous observation plus the error distribution of the corresponding quarter hour of the day.

6.2.2. Historical Sampling

The second baseline method is also based on the empiric distribution of each quarter hour of the day. In historical sampling, all values with the same quarter hour of the day are grouped together and the empirical distribution for the quarter hour is calculated. In the forecasting step, the empiric distribution of the corresponding quarter hour of the forecasting time instance is applied as the predicted distribution. In Botman, Lago, Becker, Vanthournout, and Moor [5], this simple method has shown effective results in forecasting load data up to 24 hours ahead. Notably, this method only utilizes temporal information for forecasting and does not take recent historical load data into account.

6.2.3. Linear Quantile Regression

Linear quantile regression (LinQR) is a similar estimation method to classic linear regression. The target is to find a linear relation between the predictor variable and the response variable. Consider the general linear regression problem,

$$\hat{\beta} = \arg \min_{\beta} \sum_{i \in I} c(y_i, x_i \cdot \beta).$$

Here, the aim is to find the optimal parameter column vector $\hat{\beta}$, so that the sum of a general cost function $c(\cdot)$ between the observation y_i and the linear transformation of the explanatory row vector x_i is minimized over a general index set I . In standard linear regression, the cost function $c(\cdot)$ is the mean squared error, which leads to an closed algebraic solution. However, in LinQR $c(\cdot)$ the pinball loss for a fixed quantile is taken as a cost function as defined in section 2.2.3. This results in a linear programming problem, which has to be solved algorithmically. For each quantile, a regression model has to be trained individually. However, due to the independent training of different quantiles, *quantile crossing* can happen, meaning that it possible that quantile predictions for a smaller quantile are greater than bigger quantiles. For this reason for multiple quantile regression, the predicted quantiles are sorted in ascending order.

For the benchmark model, we trained a LinQR model for each quantile in for 0.01, 0.02,..., 0.99 using the Python package scikit-learn and the HiGHS solver [23]. For the predictor variables, we applied a similar feature selection as in Botman, Lago, Becker, Vanthournout, and Moor [5]. The explanatory vector consists of historical data up to 10 time steps into the past, 2 shifted values of the exact daytime of one and two weeks prior to the forecasting time instance, and temporal data of the daytime and month of the year encoded with sine and cosine to retain the cyclic characteristic. Thus, if the load observation o_t has to be predicted, the explanatory vector is defined by

$$x_t = [o_{t-1}, o_{t-2}, \dots, o_{t-10}, o_{t-96.7}, o_{t-96.14}, \sin(\text{time}(o_t)), \cos(\text{time}(o_t)), \sin(\text{month}(o_t)), \cos(\text{month}(o_t))].$$

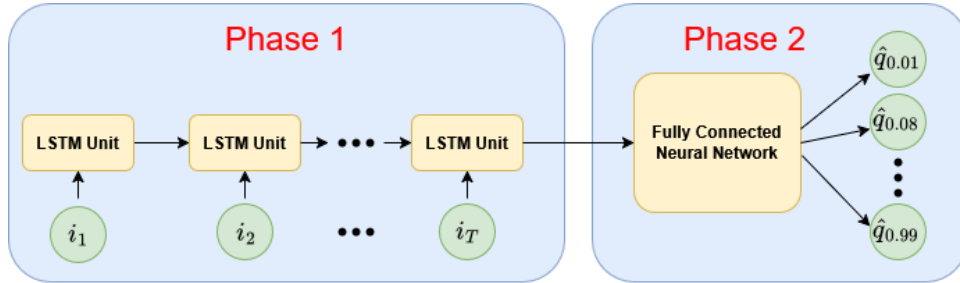


Figure 27.: Overall structure of the pinball loss guided LSTM.

6.2.4. Long short-term memory network

Long short-term memory networks (LSTM) are considered one of the most competitive neural network designs and achieve in various fields great results, also for energy forecasting. Here, we rely on an adapted model of Wang, Gan, Sun, Zhang, Lu, and Kang [36] which was further developed by Xu, Hu, and Fan [37].

The simplified model presented here consists of two phases, visualized in figure 27. In the LSTM phase, which iterates over the input time series $\{i_t\}_{t=1,\dots,T}$ to recognize relevant short- and long-term information. This information is then forwarded to a multilayered

fully-connected neural network, which utilizes the output of the LSTM phase and translates it to the desired quantile forecast vector.

For backpropagation, the pinball loss as defined in section 2.2.3 is used as the error metric. To limit the computational effort, we design the model to predict 15 quantiles reaching from 0.01% to 0.99% in 0.07% gaps and interpolate the missing quantiles linearly. For each prediction, the LSTM model receives a time series of input vectors reaching back 2 days implying the historic window length $T = 192$. The single input vectors consist of the historic load data and the temporal data equal to the one used in the LinQR, thus for $n \in \{1, \dots, T\}$

$$i_n = [o_{t-T+n-1}, \sin(\text{time}(o_{t-T+n-1})), \cos(\text{time}(o_{t-T+n-1})), \\ \sin(\text{month}(o_{t-T+n-1})), \sin(\text{month}(o_{t-T+n-1}))]$$

if the distribution of observation o_t is to be predicted. We implemented the model in Julia using the Flux package with the ADAMS optimizer set to the learning rate of 0.001 [26] and applied the hyperparameter setting described in table 7. During the training phase, the maximum number of epochs is set to 20 with a batch size of 1024.

Hyperparameter	Value
LSTM unit size	64
Neural unit 1 size	128
Neural unit 2 size	64

Table 7.: Hyperparameter setting for the LSTM model.

6.3. Results & Discussion

The results of the benchmarks are visualized in the figures 28-32. The models HMM-A and HMM-B correspond with the optimal HMM models for the different discretization types. These bar charts show the mean-CRPS of all models sorted in ascending order by best performance for each household separately, while the numerical results are summarized in table 9 in appendix C.

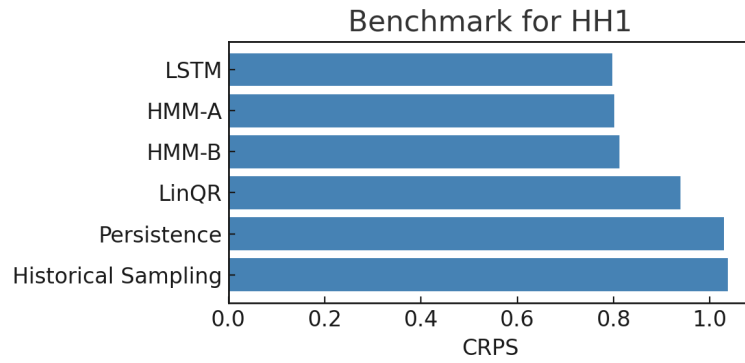


Figure 28.

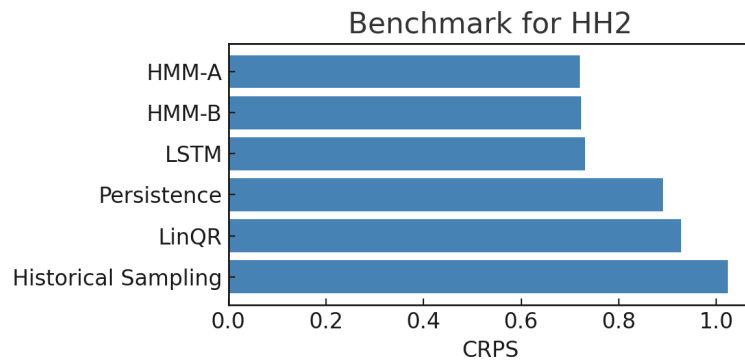


Figure 29.

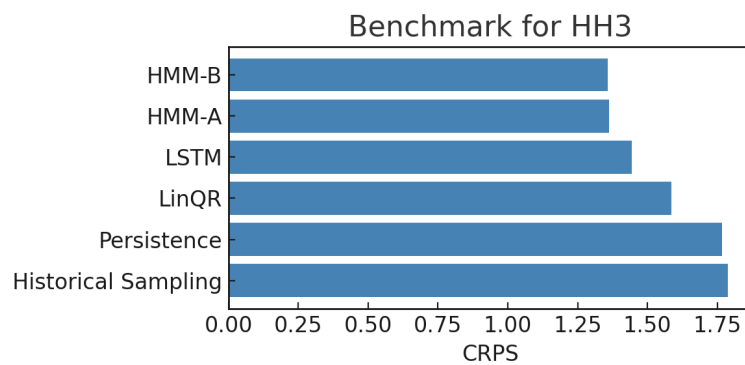


Figure 30.

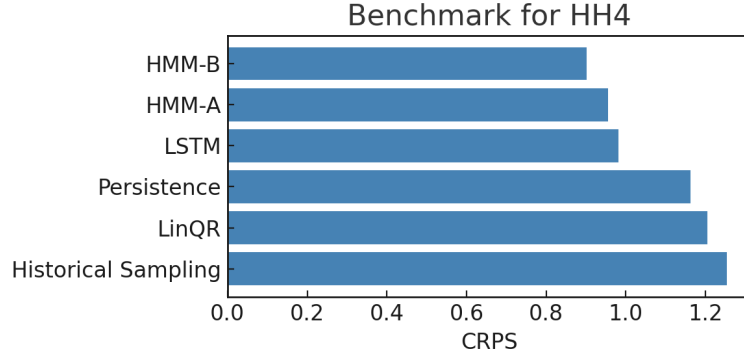


Figure 31.

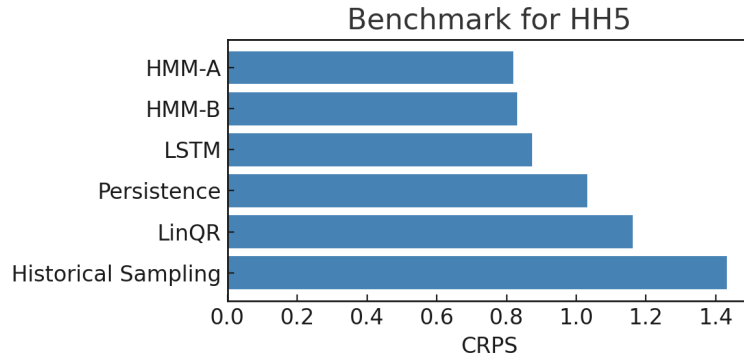


Figure 32.

The charts show, that the models with the best performance are HMM-A, HMM-B and LSTM which are always ranked above the other models. The persistence model and LinQR are located in the midfield, while historical sampling is consistently the worst model. Both HMM models generally perform slightly better than LSTM with exception of HH1 in which LSTM achieves the best mean-CRPS value by a small margin. In difference to the experiments conducted for the HMM models on the validation data set, are HMM models with discretization type B not consistently better than HMM models with discretization type A. A standard methodic of probabilistic forecasting evaluation is to calculate the relative improvements of the mean-CRPS in comparison to the baseline models. This allows a fast subsumptions of the results and comparisons between literature. Same as in Botman, Lago, Becker, Vanthournout, and Moor [5], our chosen baseline model is the persistence model to computed the relative improvements, which are presented in table 8.

The relative improvements of the HMM-A and HMM-B models lie between 17% to 22% and 18% to 23% respectively. The LSTM model has a range of 15% to 22% and lacks in some case up to 7% of improvement behind the HMM models. Based on these results, it can be confidently concluded that the HMM models are able to achieve similar and often better results in predicting the one-step-ahead distribution than the LSTM model and other benchmark models. However, to verify this result further the experiments should be repeated on a much bigger number of households. For example in Botman,

HH	Hist. Sampling	LinQR	LSTM	HMM-A	HMM-B
HH1	-0.78%	8.75%	22.45%	22.06%	21.09%
HH2	-14.94%	-4.27%	17.87%	19.10%	18.76%
HH3	-1.19%	10.25%	18.18%	22.82%	23.16%
HH4	-7.83%	-3.70%	15.58%	17.73%	22.46%
HH5	-38.66%	-12.60%	15.41%	20.64%	19.48%

Table 8.: The relative improvement of the mean-CRPS values of the models compared to the persistence model

Lago, Becker, Vanthournout, and Moor [5] 100 households for multiple data sets were investigated. In addition, it should be considered that no data set specific hyperparameter tuning was conducted for the benchmark models which would have exceeded the scope of this thesis. Thus, we expect that with a more detailed study that the benchmark models can improve slightly. However, also the HMM models could be potentially improved since in the here presented design do not make use of temporal information as it is standard for load forecasting models. This can be achieved by either adding temporal data as a additional feature to the observation state, or train multiple HMM models for different day times (like eg. night vs day) or seasons. The here inferred results are in any case promising that HMM can be considered as a competitive alternative in probabilistic load forecasting. Nevertheless, future studies to this topic are necessary.

Lastly, we want to put the results of this thesis in relation to existing literature. In Botman, Lago, Becker, Vanthournout, and Moor [5], similar benchmark models and results are obtained in a detailed study for a global approach of forecasting individual households load. In difference to this thesis, they conducted a multi-step forecasting method to predict the distributions for each quarter hour of the next 24 hours. Compared to the same baseline model, the evaluated forecasting models achieve relative improvements of 20% to 34% which is significantly. This general better performance compared to the here presented results is explained by the difference of forecasting horizon. For the one-step-ahead prediction, the persistence model is more accurate than for a longer forecasting horizon which implies that also the relative improvements are lower for the one-step-ahead model. Nevertheless, this comparison shows that the results of this thesis have a similar improvement level and thus, are in line with existing literature. Another interesting observation of this literature is that the historical sampling model performs for the 24h-multiple-step-ahead prediction equally good as the advanced forecasting method while in the one-step-ahead prediction performs very poorly. This effect is most probably caused by the fact that the historical sampling model does not take recent historic data into account but relies purely on temporal information. This implies that for very short-time forecasting, temporal data does not have a great significance as recent historic data.

7. Conclusion

This thesis contributes to the area of forecasting by presenting a novel method for probabilistic forecasting based on HMM. Although HMM has been a popular modeling method for decoding and classification, it has been applied only scarcely for prediction purposes with little to no general theoretic background. For this reason, we developed a simple theoretic framework for HMM based forecasting in align with the existing standard HMM literature and derived an efficient HMM forecasting algorithm. This algorithm can be generally used for different forecasting targets. We however, saw the strengths of the HMM method in probabilistic forecasting since the stochastic nature of the process allows an efficient probabilistic consideration of all the available information. Probabilistic forecasting acknowledges in opposite to deterministic forecasting the uncertain behavior of predictions and aims to measure and minimizes this uncertainty. We also dedicated one section of this thesis to introduce the fundamentals of probabilistic forecasting leading up to the main principle of probabilistic forecasting: "Probabilistic forecasting has the general goal of maximizing the sharpness of the predictive distributions, subject to calibration." For evaluation of probabilistic forecasts, in specific distribution forecast, we discuss one popular scoring rule namely the CRPS, which accounts for calibration and sharpness. To test the predictive capability of HMM, we selected the use-case of electrical power load forecasting of households. The electrical demand of household suits as an use-case problem for the HMM forecasting method of distribution because of two reasons. First, the household's demand for electricity underlies great variability for which probabilistic forecasting methods comprise these uncertainties. Secondly, there exists great interest in research of optimizing low-voltage level systems including forecasting methods due to the increase of challenges to the distribution net in the upcoming future. Here, the presented HMM forecasting method offers a great alternative to existing probabilistic forecasting methods due to its efficiency.

The dedicated target was to forecast the distribution of the electric power load 15 minutes ahead into the future. We tested the model on 5 different households from the WPuQ data. For the implementation of the HMM forecasting model, a detailed machine learning methodology was applied. Since we worked with a discrete state and observation HMM for a continuous regression problem, pre- and postprocessing of the data are necessary including the separation into two different discretization types. While one discretization follows standard procedure with equidistant bin splitting, another method was based on occurrences which benefits the design of the HMM model. For both of these discretization approaches, optimal models were deduced from the hyperparameter tuning. While the number of hidden states differ for the optimal models and lie between 40 and 80, the optimal number of observations is almost consistently 100 with only one model being the exception with 50 observations. A sensitivity analysis of the historic window length, which

is considered during the forecasting process, results that for most models a historic window length reaching back two and a half hours is sufficient to achieve optimal forecasting results. In a few cases, a greater historic window length was needed. These results show that for future experiments and applications of the HMM forecasting model for household electricity load data 100 observations are a good choice for the number of discretizations while for the number of hidden states a hyperparameter tuning is recommended and should focus on the area between 40 to 80 states. For the presented one-step-ahead forecasting problem, a relative short historic window length is sufficient while it also reveals limitations of the proposed forecasting method. To wit, it shows that the HMM model does not consider information prior to this historic window and implying that it cannot detect patterns or behavior happening before.

In the next experiments, we studied the optimal models in detail to gain more insights of the model predicting behavior. First, a classic evaluation of the mean-CRPS concluded that there are no major differences of performance between the two proposed discretization types. However, a closer inspection of the PIT histograms revealed that models with discretization into equidistant bins have a significant worse calibration than models with equal-mass bins. As a last study, the predicted distributions for multi-step-ahead forecasts were investigated and showed that for most cases the predicted distribution converges to the empirical distribution at a time horizon of around 5 hours into the future. We consider this time horizon as the upper limit for a potential application of the HMM forecasting method.

Lastly, we compared the HMM forecasting models with other probabilistic forecasting methods. For the benchmark experiments, two heuristic models serve as a baseline, while one statistical and one machine learning model represent the state-of-the-art methods. The results show that the HMM models outperform in most cases the competing models, and only for one household does it achieve equal results. The best benchmark model, the LSTM, accomplishes a relative improvement to the baseline reaching from 15% to 22%, while the relative improvement of the HMM models ranges from 17% to 23%. These are promising results, however, a few limitations have to be considered. Due to the scope of this thesis, no household specific hyperparameter tuning was conducted for the benchmark models. While this work served as a first study of the potential usage of HMM as a forecasting method, further experiments are needed to verify the results on a large scale. As a recommendation for future work, the model should be tested on larger data sets which are optimally already popular in the area of electrical load forecasting to allow comparability to historical and future literature. The HMM forecasting model can also be extended in various ways. One challenge is to design the HMM model to include temporal data and model typical daytime patterns. Another option is to consider a continuous observation HMM instead of the discrete version. For this case, the pre- and postprocessing step can be reduced, but certain assumptions of the distribution have to be made.

In conclusion, this thesis shows that HMM has the potential to become a competitive alternative to existing probabilistic forecasting methods, even though future research is needed to exploit several open questions.